



Chapter # 07

Skeletal System of Man



Student Learning Outcomes

At the end of this chapter, students will be able to:

- [B-12-R-61] Describe the structure of bone and compare it with that of cartilage.
- [B-12-R-62] Explain the functions of osteoblasts, osteoclasts and osteocytes.
- [B-12-R-63] Identify the main divisions of the human skeleton and list the bones of the appendicular and axial skeleton of man.
- [B-12-R-64] Describe three types of joints i.e., fibrous joints, cartilaginous joints and synovial joints and give example of each.
- [B-12-R-65] Describe the disorders of human skeleton (disc-slip, spondylosis, sciatica, arthritis, osteoporosis) and their causes.
- [B-12-R-66] State different types of fractures (simple, compound and complicated) and describe the repair process of simple fractures.
- [B-12-R-67] Describe the injuries in joints (dislocation and sprain) and their first aid treatment.
- [B-12-R-68] Compare smooth muscles, cardiac muscles and skeletal muscles.
- [B-12-R-69] Annotate the ultrastructure of the skeletal muscle.
- [B-12-R-70] Explain the sliding filaments model of muscle contraction.
- [B-12-R-71] Describe the action of antagonistic muscles in the movement of knee joint.
- [B-12-R-72] Explain muscle fatigue, cramps and tetany.
- [B-12-R-73] Differentiate between tetanus and muscle tetany.

Support is a fundamental requirement for all living organisms. It is essential to maintain body shape, resist the pull of gravity, and withstand various external forces. As organisms evolved and increased in size, their structural demands became more complex. This need for support became particularly critical when animals transitioned from aquatic to terrestrial environments, where buoyant forces no longer assisted in maintaining body structure. In animals, this vital supportive function is primarily carried out by the skeleton, which provides a rigid framework for maintaining posture, facilitating movement, and protecting internal organs.

The human skeletal system, or human skeleton, is a highly organized structure composed of bones and cartilage. Bones form the primary rigid framework, while cartilage provides flexible support at specific points such as joints. Together, these components perform several essential functions. They provide structural support to the body's soft tissues, enabling posture and stability. They facilitate movement by serving as attachment sites for skeletal muscles, which act across joints to produce coordinated motion. In addition to support and locomotion, the skeletal system protects vital organs, stores essential minerals such as calcium and phosphorus, and houses bone marrow, which is responsible for the formation of blood cells.

Thus, the human skeleton is not merely a static framework but a dynamic system integral to movement, protection, and overall physiological function, reflecting the remarkable adaptability and complexity of the human body.



STRUCTURE OF BONE AND COMPARISON WITH CARTILAGE

1. Structure of Bone

Bone is a highly specialized, rigid connective tissue that forms the primary component of the human skeleton. It provides structural support, facilitates movement, protects vital organs, stores minerals, and houses bone marrow. Bone tissue exhibits both strength and resilience due to its unique composition and organization.

The human bone is a remarkable composite material, combining both organic and inorganic components to achieve a balance of strength, flexibility, and resilience. This unique composition allows bones to withstand mechanical stresses while maintaining a lightweight structure essential for movement.

► Composition of Bone

(a) Organic Component (≈30–35%)

The organic portion of bone is primarily responsible for its flexibility and tensile strength, preventing bones from being brittle under stress. It consists of:

- **Collagen fibers:** Type I collagen is the dominant protein in bone. These long, fibrous molecules form a dense, interwoven network that provides tensile strength, allowing bones to resist stretching and bending forces. Collagen fibers also serve as a scaffold for the deposition of minerals during bone formation.
- **Non-collagenous proteins:** These include osteocalcin, osteonectin, and bone sialoprotein. They play critical roles in regulating mineralization, binding calcium ions, and facilitating the proper organization of hydroxyapatite crystals within the bone matrix. Additionally, they contribute to bone repair and remodeling, ensuring that micro-damage from daily stress is continually repaired.
- **Proteoglycans and glycoproteins:** These molecules, though present in smaller amounts, retain water within the bone matrix, adding slight compressive resilience and contributing to the elasticity of bone.

(b) Inorganic Component (≈65–70%)

The inorganic fraction is responsible for the hardness and compressive strength of bone. It consists mainly of:

- **Hydroxyapatite [$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$]:** This calcium phosphate mineral is deposited in and around the collagen framework. Hydroxyapatite crystals form a rigid lattice that makes bones hard and capable of supporting body weight and resisting mechanical pressure.
- **Other mineral ions:** Small amounts of magnesium, sodium, carbonate, and fluoride ions are also incorporated into the bone matrix. These ions influence the crystallinity, solubility, and overall strength of the mineralized bone, and help in maintaining mineral homeostasis in the body.
- **Function of inorganic components:** By providing rigidity, the inorganic component allows bones to act as structural supports and levers for muscle action. The mineralized matrix also serves as a reservoir for essential ions like calcium and phosphate, which can be mobilized when needed for physiological functions such as nerve conduction, muscle contraction, and blood clotting.

► Integration of Organic and Inorganic Components:

The interplay between the organic collagen matrix and the inorganic hydroxyapatite crystals gives bone its unique combination of strength and toughness. Collagen fibers absorb tension, while hydroxyapatite resists compression. This composite nature ensures that bone is neither too brittle (like pure mineral) nor too flexible (like cartilage), making it ideal for the mechanical demands of the human body.

► Microscopic Structure of Bone

Bone is not a homogeneous tissue but exhibits a highly organized microscopic architecture that enables it to fulfill multiple structural and physiological functions. At the microscopic level, bone can be broadly categorized into **compact (cortical) bone** and **spongy (cancellous or trabecular) bone**, each with distinct structural features and functional significance.

(a) Compact (Cortical) Bone

Compact bone forms the dense, outer layer of bones, providing strength and rigidity. It is most prominent in the diaphysis (shaft) of long bones. The key structural unit of compact bone is the **osteon**, also known as the Haversian system.

- **Osteons:** Each osteon consists of a central **Haversian canal** that contains blood vessels, nerves, and lymphatic vessels, ensuring the supply of nutrients and removal of metabolic waste. Surrounding the canal are concentric layers of mineralized matrix called **lamellae**, which are organized in rings to resist torsional and compressive forces.
- **Osteocytes and lacunae:** **Osteocytes**, the mature bone cells, are embedded within small cavities called **lacunae** situated between the lamellae. These cells maintain the bone matrix and communicate with each other through fine, microscopic channels known as **canaliculi**. Canaliculi form an extensive network that allows the exchange of nutrients, gases, and metabolic wastes between osteocytes and the blood vessels in the Haversian canals.
- **Interconnection:** Osteons are interconnected by **Volkman's canals**, which run perpendicular to the Haversian canals, linking adjacent osteons and integrating the vascular network throughout the compact bone. This organization ensures that compact bone is highly resilient and capable of bearing significant mechanical loads.

(b) Spongy (Cancellous or Trabecular) Bone

Spongy bone is found primarily at the **epiphyses** of long bones, within flat bones, and surrounding the medullary cavity. Unlike compact bone, it is less dense and has a porous, lattice-like arrangement that maximizes strength while minimizing weight.

- **Trabeculae:** The porous structure consists of thin plates and rods called **trabeculae**, which are oriented along lines of mechanical stress. This arrangement allows spongy bone to resist multidirectional forces efficiently while keeping the bone lightweight.
- **Bone marrow spaces:** The spaces between trabeculae are filled with **red bone marrow**, the site of hematopoiesis (formation of red blood cells, white blood cells, and platelets). This feature makes spongy bone crucial not only for structural support but also for maintaining vital physiological processes.
- **Functional advantage:** The trabecular architecture also provides slight flexibility, reducing the risk of fractures by absorbing shocks and distributing stress throughout the bone.

► Structural Parts and Internal Composition of Long Bone

Structurally, a **long bone** consists of two expanded ends called **epiphyses** and a central shaft known as the **diaphysis**. The diaphysis encloses a **medullary cavity**, which is usually filled with **yellow bone marrow**, a tissue primarily involved in fat storage.

- **Periosteum:** The outer surface of the bone is enveloped by the **periosteum**, a tough fibrous membrane that serves as an anchoring site for tendons and ligaments. It contains blood vessels, lymphatic vessels, and nerves, all of which contribute to the nourishment and repair of bone.
- **Endosteum:** The inner surfaces of bone, including the medullary cavity and spaces within spongy bone, are lined by the **endosteum**, a thin vascular membrane rich in osteogenic cells, which plays a key role in bone growth, repair, and remodeling.

This gross structure not only reflects the mechanical strength of bone but also its adaptability and role in energy storage and blood cell formation.

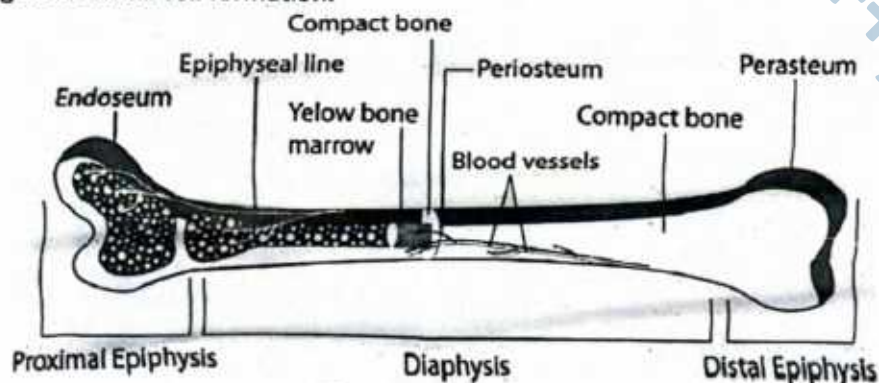


Figure: Structure of bone

► Cellular Components of Bone

Bone is a living tissue composed of specialized cells that work in a coordinated manner to maintain its structure, enable growth, and regulate mineral balance. The cellular components of bone can be classified into three main types: **osteoblasts, osteocytes, and osteoclasts**. Each plays a distinct and crucial role in bone physiology.

(a) Osteoblasts – The Bone-Forming Cells

Osteoblasts are mononuclear, cuboidal cells located on the surface of developing or remodeling bone. They are primarily responsible for the **synthesis of the organic bone matrix**, known as **osteoid**, which is rich in collagen fibers (mainly type I) and non-collagenous proteins. Osteoblasts initiate the process of **mineralization**, depositing hydroxyapatite crystals within the osteoid to harden the bone.

Functions of Osteoblasts:

- Formation of new bone tissue during growth, repair, and remodeling.
- Secretion of signaling molecules that regulate osteoclast activity and overall bone homeostasis.
- Participation in hormonal regulation of calcium and phosphate metabolism.

(b) Osteocytes – The Mature Bone Cells

Osteocytes are former osteoblasts that have become **embedded within the mineralized bone matrix**. They reside in small spaces called **lacunae** and extend long cytoplasmic processes through **canaliculi**, forming an extensive intercellular network.

Functions of Osteocytes:

- Maintenance of bone tissue by regulating the deposition and resorption of minerals in their immediate environment.
- Sensing mechanical stress and strain on bone, thereby directing remodeling to strengthen areas experiencing higher loads.
- Communication with osteoblasts and osteoclasts to coordinate bone formation and resorption.

(c) Osteoclasts – The Bone-Resorbing Cells

Osteoclasts are **large, multinucleated cells** derived from the fusion of monocyte/macrophage precursors. They are primarily involved in **bone resorption**, a process in which the mineralized matrix is broken down, releasing calcium and phosphate ions into the bloodstream.

Functions of Osteoclasts:

- Remodeling of bone to adapt to mechanical stress and repair micro-damage.
- Regulation of blood calcium and phosphate levels, essential for physiological functions such as nerve conduction and muscle contraction.
- Working in tandem with osteoblasts to maintain a dynamic balance between bone formation and resorption, a process known as **bone remodeling**.

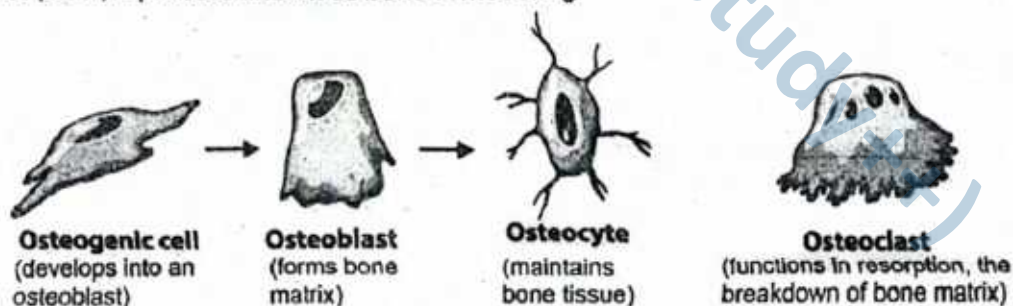


Figure: Types of bone cells

► Integration of Cellular Functions

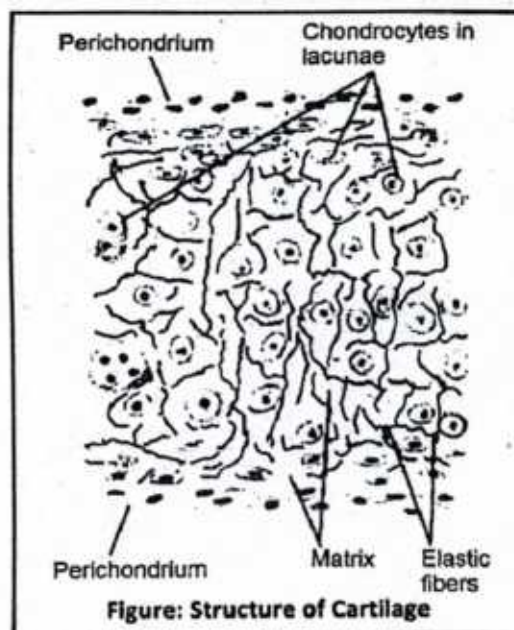
The coordinated activity of osteoblasts, osteocytes, and osteoclasts ensures that bone is a **dynamic tissue**—capable of growth, repair, and adaptation throughout life. Osteoblasts build new bone, osteocytes monitor and maintain it, and osteoclasts resorb old or damaged bone. This cellular interplay maintains skeletal strength, facilitates calcium homeostasis, and allows the skeleton to adapt to changing mechanical demands.

► Gross Structure:

- Long bones consist of a **diaphysis** (shaft), **epiphyses** (ends), and **metaphysis** (region between shaft and ends).
- Articular surfaces are covered with a thin layer of cartilage to reduce friction.

2. Structure of Cartilage

Cartilage is a specialized form of connective tissue that combines strength with flexibility. It provides support to various body parts, maintains structural integrity, and ensures smooth movements at joints by reducing friction and absorbing mechanical shocks. Unlike bone, cartilage is **avascular**—it lacks blood vessels, lymphatics, and nerves. Instead, nutrients and oxygen diffuse through the matrix from surrounding tissues. This limited nutrient supply makes cartilage growth and repair much slower than bone, contributing to the persistence of cartilage injuries.



► Composition of Cartilage

Cartilage is composed of **cells** and **extracellular matrix**, both of which are adapted for its supportive and cushioning roles.

1. Cells:

- **Chondroblasts:** These are the immature, metabolically active cells located near the periphery of cartilage. They secrete the extracellular matrix, including collagen fibers, proteoglycans, and glycoproteins. Chondroblasts play a crucial role during cartilage development and growth.
- **Chondrocytes:** These are mature cartilage cells derived from chondroblasts. They are embedded within small spaces called **lacunae** inside the matrix. Chondrocytes are responsible for maintaining the cartilage matrix, regulating its turnover, and ensuring structural stability.

2. Extracellular Matrix:

- **Fibrous component:** Rich in **type II collagen fibers**, which provide tensile strength and prevent tearing under stress. In fibrocartilage, type I collagen is also abundant, contributing to its exceptional toughness.
- **Ground substance:** Contains **proteoglycans** (such as aggrecan) and **glycosaminoglycans** (like chondroitin sulfate and hyaluronic acid), which trap large amounts of water. This hydrated gel-like consistency makes cartilage resistant to compression and ideal for absorbing shock.
- **Water content:** Cartilage matrix is up to **80% water**, allowing it to be resilient, *elastic*, and compressible. The high water content also facilitates the diffusion of nutrients and metabolic wastes.

► Types of Cartilage

Cartilage exists in three specialized forms, each adapted for different structural and functional roles:

1. Hyaline Cartilage (Glassy cartilage):

- **Features:** The most abundant type, with a smooth, translucent matrix that appears glassy under the microscope. It has a moderate amount of collagen fibers and a relatively high water content.
- **Locations:** Found at articular surfaces of long bones, ends of ribs, nasal septum, larynx, trachea, and bronchi.
- **Functions:** Provides a smooth, low-friction surface for joint movement; supports respiratory airways; and contributes to the structural framework of the nose and rib cage.

2. Elastic Cartilage:

- **Features:** Contains abundant **elastic fibers** interwoven with collagen, making it flexible and resilient.
- **Locations:** Found in the external ear (pinna), epiglottis, and auditory (Eustachian) tube.
- **Functions:** Provides flexible support while maintaining shape; allows bending without permanent deformation.

3. Fibrocartilage:

- **Features:** Contains dense bundles of **type I collagen fibers** arranged in parallel, giving it exceptional tensile strength. It lacks a perichondrium (outer covering), making it less capable of repair.
- **Locations:** Found in intervertebral discs, pubic symphysis, and menisci of the knee joint.
- **Functions:** Acts as a tough shock absorber; resists both compression and tensile forces; provides strong support in regions subject to heavy pressure and stress.



► **Key Characteristics of Cartilage**

- **Avascularity:** No blood vessels; nutrients supplied by diffusion through the matrix.
- **Growth:** Can grow by two methods:
 - **Interstitial growth:** Division of chondrocytes within the matrix, expanding cartilage from within (important in early development).
 - **Appositional growth:** Addition of new layers by chondroblasts in the perichondrium (outer layer of cartilage).
- **Repair:** Limited due to absence of direct blood supply; injuries often heal slowly or incompletely.

3. Comparison between Bone and Cartilage

Feature	Bone	Cartilage
Matrix	Mineralized (rigid, contains hydroxyapatite)	Non-mineralized, gelatinous (flexible)
Cells	Osteoblasts, osteocytes, osteoclasts	Chondroblasts, chondrocytes
Vascularity	Highly vascular	Avascular (nutrients by diffusion)
Growth	Both appositional and interstitial, mainly at epiphyseal plates	Interstitial and appositional growth
Strength & Flexibility	Hard, strong, resists compression	Flexible, resists compression and tension
Repair Ability	Rapid due to blood supply	Slow due to lack of blood vessels
Location	Skeleton, supporting framework, protection	Joints, nose, ear, intervertebral discs

Bone and cartilage are complementary connective tissues in the human skeletal system. Bone provides rigid support, mineral storage, and protection, whereas cartilage offers flexibility, cushioning, and smooth surfaces at articulating joints. The combination of these tissues ensures structural integrity, mobility, and resilience in the human body.

SLO [B-12-R-62]

Explain the functions of osteoblasts, osteoclasts and osteocytes.

FUNCTIONS OF OSTEOBLASTS, OSTEOCLASTS, AND OSTEOCYTES

Bone is a dynamic tissue that constantly undergoes **remodeling**, a process that involves the *balanced* activity of three major cell types: **osteoblasts**, **osteoclasts**, and **osteocytes**. Their coordinated functions ensure growth, repair, and mineral homeostasis throughout life.

1. Osteoblasts – The Bone-Building Cells

- **Function in bone formation:** Osteoblasts secrete the organic matrix (osteoid), which is composed primarily of type I collagen and non-collagenous proteins. They also initiate the deposition of calcium phosphate crystals, leading to the mineralization and hardening of bone.
- **Role in growth and repair:** Osteoblasts are active during bone growth, fracture healing, and remodeling. They form new bone tissue on surfaces where additional strength is needed.
- **Endocrine function:** Osteoblasts release signaling molecules such as osteocalcin, which not only regulate bone mineralization but also influence energy metabolism and glucose regulation.
- **Fate of osteoblasts:** Once their bone-forming activity declines, they may become inactive **bone-lining** cells, undergo apoptosis, or differentiate into osteocytes.

2. Osteocytes – The Bone-Maintaining Cells

- **Function in maintenance:** Osteocytes are embedded within lacunae of the bone matrix and maintain the surrounding mineralized tissue by regulating ion exchange between bone and blood.
- **Role as mechanosensors:** Osteocytes detect mechanical stress and strain placed on bone. Through their cytoplasmic extensions in canaliculi, they signal osteoblasts and osteoclasts to strengthen or resorb bone in response to changing loads.
- **Coordination of remodeling:** By releasing chemical messengers, osteocytes act as the "command center" of bone tissue, directing remodeling processes to ensure skeletal strength and integrity.
- **Nutrient and waste exchange:** Their canalicular network allows exchange of nutrients and removal of waste products, ensuring survival in the mineralized matrix.



3. Osteoclasts – The Bone-Resorbing Cells

- **Function in bone resorption:** Osteoclasts secrete acids and enzymes that dissolve the inorganic mineral and digest the organic collagen matrix, breaking down old or damaged bone.
- **Role in remodeling:** By removing weakened bone, osteoclasts create space for osteoblasts to deposit new matrix, thereby maintaining skeletal strength.
- **Calcium regulation:** Osteoclast activity releases calcium and phosphate ions into the bloodstream, playing a critical role in maintaining mineral balance for essential physiological functions such as nerve impulse conduction, muscle contraction, and blood clotting.
- **Repair and adaptation:** Osteoclasts work in concert with osteoblasts to reshape bone according to mechanical stresses, ensuring bones remain efficient structures without unnecessary weight.

► Integration of Functions

The coordinated actions of osteoblasts, osteocytes, and osteoclasts ensure that bone remains a **living, adaptable tissue**:

- **Osteoblasts** build new bone.
- **Osteocytes** monitor and maintain the existing bone matrix.
- **Osteoclasts** resorb bone, allowing remodeling and mineral release.

Together, they maintain skeletal integrity, enable growth and repair, and regulate calcium homeostasis—functions that are essential for overall health and survival.

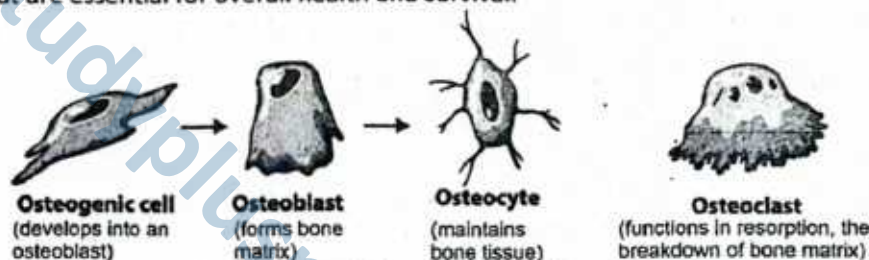


Figure: Types of bone cells

Feature	Osteoblasts	Osteocytes	Osteoclasts
Origin	Derived from osteogenic progenitor cells	Derived from osteoblasts	Derived from monocyte / macrophage lineage
Location	Found on the surface of bone tissue	Reside in lacunae within the bone matrix	Found on bone surfaces in resorption bays
Function	Synthesize and secrete unmineralized bone matrix (osteoid)	Maintain bone tissue and regulate mineral content	Break down bone tissue (bone resorption)
Activity	Active during bone growth and repair	Metabolically less active; communicate via canaliculi	Highly active during bone remodeling and calcium regulation
Nucleus	Single, centrally located	Single, centrally located	Multinucleated

SLOs Based (MCQs)

- Which of the following bone cells is responsible for bone formation?
 - Osteocyte
 - Osteoblast
 - Osteoclast
 - Chondrocyte
- Osteocytes are located in:
 - Periosteum
 - Haversian canal
 - Lacunae within the bone matrix
 - Perichondrium
- The function of osteoclasts is to:
 - Maintain bone matrix
 - Form new bone
 - Break down bone tissue
 - Produce cartilage
- The matrix of cartilage is primarily:
 - Rigid and highly mineralized
 - Flexible with elastic and collagen fibers
 - Composed of calcium phosphate only
 - Made of adipose tissue
- The outer fibrous covering of cartilage is called:
 - Periosteum
 - Epimysium
 - Perichondrium
 - Endosteum
- Which cartilage type is most abundant in the human body?
 - Fibrocartilage
 - Elastic cartilage
 - Hyaline cartilage
 - Calcified cartilage

7. Which of the following statements about bone is correct?
A. Bone is avascular and heals slowly.
B. Bone matrix contains only elastic fibers.
C. Bone is highly vascularized and heals quickly.
D. Bone does not contain living cells.
8. Chondrocytes obtain nutrients through:
A. Direct blood supply
B. Active transport from the bone
C. Diffusion through the cartilage matrix
D. Canaliculi
9. Which type of cartilage is found in intervertebral discs?
A. Hyaline cartilage B. Fibrocartilage
C. Elastic cartilage D. Calcified cartilage
10. The mineral primarily responsible for bone hardness is:
A. Sodium chloride B. Calcium phosphate
C. Magnesium sulfate D. Potassium carbonate
11. Which cartilage type contains both collagen and a high number of elastic fibers?
A. Hyaline cartilage B. Elastic cartilage
C. Fibrocartilage D. Calcified cartilage
12. Osteology refers to the:
A. Study of joints
B. Study of cartilage only
C. Study of muscles
D. Study of bones and cartilage
13. The function of osteocytes includes all of the following except:
A. Maintaining bone tissue
B. Regulating calcium release
C. Bone matrix resorption
D. Secreting enzymes
14. Cartilage is avascular. Which of the following is a consequence of this?
A. Cartilage is easily broken
B. Cartilage resists compression
C. Cartilage heals slowly
D. Cartilage becomes rigid with age
15. Periosteum is:
A. A vascular membrane lining the bone marrow cavity
B. A protective covering around cartilage
C. The outer fibrous layer of a bone
D. A cavity where osteocytes reside
16. Which of the following pairs is correctly matched?
A. Elastic cartilage – intervertebral disc
B. Fibrocartilage – ear pinna
C. Hyaline cartilage – ends of long bones
D. Hyaline cartilage – meniscus of knee
17. The concentric arrangement of lacunae around a central canal is found in:
A. Cartilage only B. Spongy bone
C. Compact bone D. All connective tissues
18. Cartilage cells are called:
A. Osteocytes B. Fibrocytes
C. Chondrocytes D. Leukocytes
19. Which feature is common to both bone and cartilage?
A. Rich blood supply B. Presence of lacunae
C. Concentric lamellae D. High mineral content
20. Which cartilage type is found in the epiglottis?
A. Hyaline B. Elastic
C. Fibrocartilage D. Articular

Answer Key

1. B	2. C	3. C	4. B	5. C	6. C	7. C	8. C	9. B	10. B
11. B	12. D	13. C	14. C	15. C	16. C	17. C	18. C	19. B	20. B

Short Questions

Q.1 What is the function of osteoblasts in bone tissue?

Ans. Osteoblasts are specialized bone-forming cells found on the surface of bone tissue. They secrete **osteoid**, an organic matrix made of collagen and proteins, which later becomes mineralized with calcium and phosphate salts. Osteoblasts are highly active during **bone growth**, **fracture repair**, and **remodeling**. They also regulate mineral deposition, helping bones gain strength and rigidity. Once surrounded by the matrix, many osteoblasts differentiate into osteocytes. Overall, osteoblasts ensure the continuous renewal and development of healthy bone.

MARKING SCHEME (3 Marks)

- 1 mark each for three main functions

Q.2 How do osteocytes help in maintaining bone health?

Ans. Osteocytes are mature bone cells that originate from osteoblasts and reside in small spaces called **lacunae**. They extend long cytoplasmic processes through tiny canals (canaliculi), forming a communication

MARKING SCHEME (3 Marks)

- 1 mark each for three main functions

network within bone. Their functions include: maintaining mineral homeostasis by regulating calcium and phosphate exchange between bone and blood, secreting enzymes to remodel surrounding matrix, and sensing mechanical stress to signal when bone strengthening is required. Thus, osteocytes act as the "caretakers" of bone, maintaining its strength and integrity over time.

Q.3 Describe the role of osteoclasts in bone metabolism.

Ans. Osteoclasts are large, multinucleated cells derived from monocytes and located on bone surfaces. They specialize in **bone resorption**, breaking down bone tissue to release calcium and phosphate into the bloodstream. They achieve this by secreting **enzymes and hydrochloric acid**, which dissolve the organic matrix and mineral salts. Their activity is vital in **bone remodeling**, balancing bone formation by osteoblasts, and ensuring bones adapt to stress or injury. Without osteoclasts, bones would become abnormally dense, fragile, and unable to remodel properly.

MARKING SCHEME (3 Marks)

- 1 mark each for three main functions

Q.4 Why does cartilage heal more slowly than bone?

Ans. Cartilage heals slowly because it is **avascular**, meaning it lacks direct blood vessels. Instead, nutrients and oxygen must diffuse through the dense matrix to reach chondrocytes (cartilage cells). This diffusion is slow and limits the rate of **cell division and repair**. In contrast, bone is richly supplied with blood vessels, allowing rapid delivery of nutrients, growth factors, and immune cells for healing. As a result, fractured bones may heal in weeks to months, whereas cartilage injuries, such as those in joints, often take years or remain permanent.

Q.5 What is the perichondrium, and what is its function?

Ans. The **perichondrium** is a dense connective tissue sheath surrounding most types of cartilage (except articular cartilage and fibrocartilage). It consists of two layers: an outer fibrous layer containing collagen and blood vessels, and an inner chondrogenic layer with progenitor cells. Functions include: (1) **nourishing cartilage**, since blood vessels in the perichondrium supply nutrients by diffusion, (2) **supporting growth** through chondrogenic cells that can differentiate into chondroblasts, and (3) **aiding repair** by providing new cells after injury. Thus, the perichondrium is essential for cartilage survival and regeneration.

MARKING SCHEME (3 Marks)

- Definition 1 mark, functions 2 marks

Q.6 Differentiate between the mineral content of bone and cartilage.

Ans. Bone is a rigid connective tissue with **high mineral content**, primarily **calcium phosphate (hydroxyapatite crystals)** and calcium carbonate. These minerals give bone its **hardness, compressive strength, and weight-bearing capacity**. This mineralization also makes bone an important reservoir for calcium and phosphate in the body. In contrast, cartilage has a **low mineral content**. Its matrix is mostly made up of **water, proteoglycans, and collagen fibers**, giving it a **gel-like, resilient, and flexible structure**. The absence of significant mineral salts makes cartilage softer and better suited for cushioning and flexibility rather than rigidity.

MARKING SCHEME (3 Marks)

- Mineral content of bone 1.5 marks; mineral content of cartilage 1.5 marks

Q.7 Describe the structural differences between compact and spongy bone.

Ans. **Compact bone** (dense bone) is the hard, outer layer of bone. It is organized into **osteons or Haversian systems**, with concentric lamellae surrounding central canals containing blood vessels and nerves. This dense structure makes compact bone strong and ideal for providing protection and support.

MARKING SCHEME (3 Marks)

- Compact bone structure 1.5 marks; spongy bone structure 1.5 marks

Spongy bone (cancellous bone), on the other hand, is lightweight and **porous**. It consists of a **network of trabeculae**, leaving spaces that are often filled with **red bone marrow**, where blood cell formation occurs. Its structure **reduces overall bone weight** while maintaining strength, making it particularly important in the ends of **long bones** and **flat bones**.

Q.8 State the main locations and functions of hyaline cartilage.

Ans. **Location:** **Hyaline cartilage** is the most common type of cartilage. It is found at the **ends of long bones (articular cartilage)**, in the **nose, larynx, trachea, and bronchi**, and forms the **embryonic skeleton** before bone development.

MARKING SCHEME (3 Marks)

- Location 1 mark; functions 2 marks



Functions: It provides **smooth surfaces** at joints, reducing friction and enabling easy movement. It also supports **respiratory structures** like the trachea and bronchi, preventing collapse, and forms a **template for bone formation** in early development. Its balance of resilience and strength makes it vital for both structural support and smooth joint movement.

Q.9 What are the functional roles of fibrocartilage and where is it found?

Ans. Location: Fibrocartilage is found in **intervertebral discs**, the **pubic symphysis**, and the **menisci of the knee joint**. These are areas subjected to strong pressure and compressive forces.

Functions: Fibrocartilage contains **thick bundles of collagen fibers**, making it the **toughest type of cartilage**. It provides **shock absorption**, resists both **compression and tensile stresses**, and helps stabilize joints. For example, intervertebral discs cushion the vertebrae, and menisci in the knee distribute body weight evenly across the joint. Its unique combination of strength and resilience makes it ideal for high-stress regions.

MARKING SCHEME (3 Marks)

- Location 2 marks; function 1 mark

Q.10 Compare the flexibility and location of elastic cartilage with other types.

Ans. Flexibility: Elastic cartilage is the most **flexible type of cartilage** because its matrix contains abundant **elastic fibers** in addition to collagen. This allows it to bend repeatedly without losing shape, unlike hyaline or fibrocartilage, which are less elastic.

Location: Elastic cartilage is found in body parts that require **flexibility and shape retention**, such as the **external ear (pinna)**, **epiglottis**, and **auditory (Eustachian) tube**. These locations demand a structure that can withstand movement and bending while maintaining its original form. This sets it apart from hyaline cartilage (more rigid) and fibrocartilage (more supportive).

MARKING SCHEME (3 Marks)

- Location 2 marks; flexibility 1 mark

SLO [B-12-R-63]

Identify the main divisions of the human skeleton and list the bones of the appendicular and axial skeleton of man.

DIVISIONS OF THE HUMAN SKELETON

The human skeletal system consists of **206 bones**, which are structurally organized into two major divisions:

- (1) **Axial Skeleton** (2) **Appendicular Skeleton**

This division reflects the dual functions of the skeleton: **stability and protection of vital organs** (axial skeleton), and **facilitation of movement through the limbs** (appendicular skeleton).

1. Axial Skeleton

The **axial skeleton** forms the central axis of the body and includes **80 bones**. It provides **protection for the brain, spinal cord, and organs within the thoracic cavity**. The axial skeleton consists of three major regions:

► Skull

- a) **Vertebral column** b) **Thoracic cage**

(i) Skull (29 bones):

The skull is a complex structure that protects the brain and forms the framework of the face. It is categorized into four groups:

- **Cranial bones (8):** Form the cranium, which encases and protects the brain.
 - **Unpaired bones:** Frontal, Occipital, Sphenoid, Ethmoid
 - **Paired bones:** Parietal (2), Temporal (2).
- **Facial bones (14):** Form the facial structure and provide attachment sites for muscles of expression and mastication.
 - **Paired bones (6 pairs):** Maxillae, Zygomatic, Nasal, Lacrimal, Inferior nasal conchae, Palatine
 - **Unpaired bones:** Mandible, Vomer
- **Ear ossicles (6):** The three tiny bones of the middle ear in each side, the smallest bones in the body — **Malleus, Incus, Stapes**.
- **Hyoid bone (1):** A unique, U-shaped bone located below the tongue. Unlike other bones, it **does not articulate with any other bone**. It provides attachment for tongue and neck muscles, aiding in swallowing and speech.



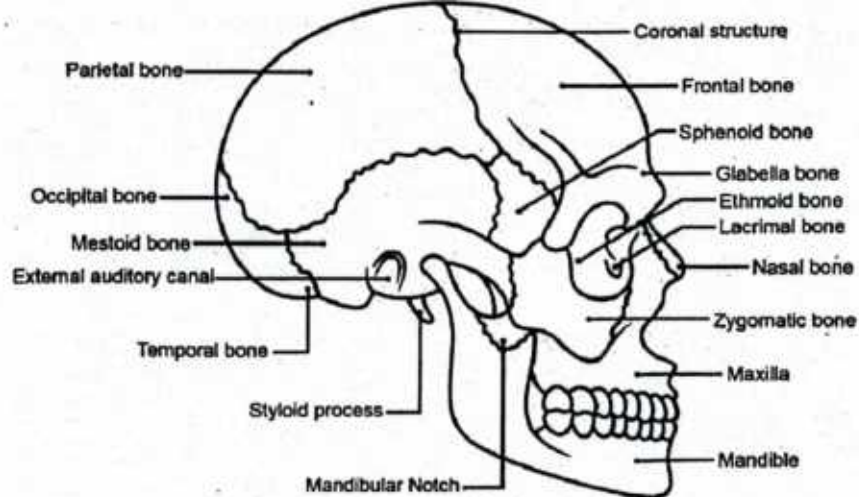


Figure: Human Skull (side view)

(ii) Vertebral Column (26 bones):

The vertebral column extends from the skull to the pelvis, forming the backbone. It protects the spinal cord and provides structural support.

Vertebral Column:

The vertebral column comprises **33 vertebrae**, arranged into five distinct regions:

Region	Number of Vertebrae	Description
Cervical	7	Vertebrae of the neck (includes atlas and axis)
Thoracic	12	Rib-bearing vertebrae of the chest
Lumbar	5	Vertebrae of the lower back
Sacral	5 (fused)	Fuse to form the sacrum
Coccygeal	4 (fused)	Fuse to form the coccyx (tailbone)

- **Atlas (C1)** is the first cervical vertebra, supporting the skull.
- **Axis (C2)** is the second cervical vertebra, allowing rotation of the head.
- The **sacrum** articulates with the iliac bones of the pelvis.
- **Sacral and coccygeal vertebrae** collectively form the **pelvic vertebrae**.

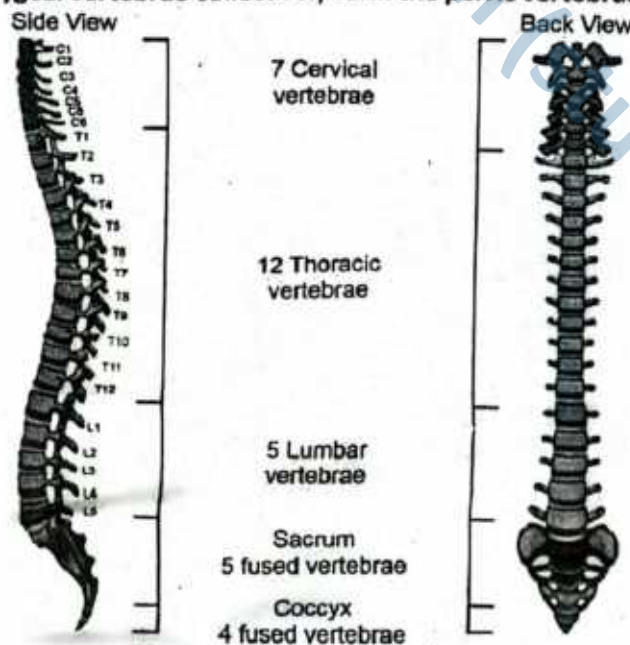


Figure: Human vertebral column

(iii) Thoracic Cage (25 bones):

The thoracic cage protects the heart and lungs while supporting respiration.

- Sternum (1)
- Ribs (24; 12 pairs)

► Rib cage

The rib cage consists of **12 pairs of ribs**, all of which articulate posteriorly with the thoracic vertebrae. The rib pairs are categorized as follows:

- **True ribs (1st–7th pairs):** Directly connected to the sternum via individual costal cartilages.
- **False ribs (8th–10th pairs):** Connected to the sternum indirectly through a shared costal cartilage.
- **Floating ribs (11th–12th pairs):** Not connected to the sternum at all.

The rib cage encloses and protects vital organs and forms the **chest cavity**, a semi-vacuum chamber essential for respiration.

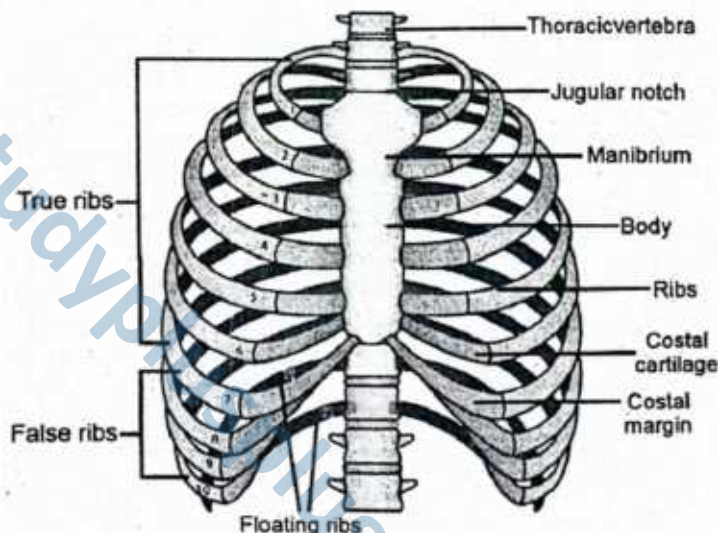


Figure: Human rib cage

2. Appendicular Skeleton

The **appendicular skeleton** consists of **126 bones** that form the limbs and their supporting girdles. It enables locomotion and manipulation of the environment.

(i) Pectoral (Shoulder) Girdle (4 bones):

- Clavicle (2) (collar bones): Each articulates medially with the sternum and laterally with the scapula.
- Scapula (2) (shoulder blades): Broad, flat bones that provide muscle attachment sites.

(ii) Upper Limbs (60 bones):

- **Arm:** Humerus (2)
 - **Humerus:** The long bone of the upper arm. Its head fits into the glenoid cavity of the scapula forming the shoulder joint.
- **Forearm:** Radius (2), Ulna (2)
 - **Radius:** The outer bone of the forearm on the thumb side.
 - **Ulna:** The inner bone of the forearm, slightly longer than the radius.
- **Hand:** Carpals (16), Metacarpals (10), Phalanges (28)
 - **Carpals:** Eight small bones arranged in two rows, forming the wrist.
 - **Metacarpals:** Five bones forming the palm.
 - **Phalanges:** Bones of the fingers. Each finger has three phalanges (proximal, middle, distal), except the thumb, which has two.

(iii) Pelvic Girdle (2 bones):

- **Coxal (hip) bones (2)** — each formed by fusion of ilium, ischium, and pubis
The two hip bones meet anteriorly at the **pubic symphysis** and articulate posteriorly with the sacrum. A deep socket called the **acetabulum** accommodates the head of the femur.

(iv) Lower Limbs (60 bones):

- **Thigh: Femur (2)**
 - **Femur:** The longest and strongest bone of the body; articulates with the acetabulum to form the hip joint.
- **Kneecap: Patella (2)**
 - **Patella:** A triangular sesamoid bone (kneecap) embedded in the quadriceps tendon.
- **Leg: Tibia (2), Fibula (2)**
 - **Tibia:** The larger, weight-bearing bone of the lower leg (shin bone).
 - **Fibula:** A thinner bone lateral to the tibia; provides muscle attachment but bears little weight.
- **Foot: Tarsals (14), Metatarsals (10), Phalanges (28)**
 - **Tarsals:** Seven tightly connected bones forming the ankle.
 - **Metatarsals:** Five bones forming the sole of the foot.
 - **Phalanges:** Bones of the toes. Each toe has three phalanges, except the big toe, which has two.

Divisions of Human Skeleton

Division	Number of Bones	Major Components
Axial Skeleton	80	Skull (29), Hyoid (1), Ear ossicles (6), Vertebral column (26), Thoracic cage (25)
Appendicular Skeleton	126	Pectoral girdle (4), Upper limbs (60), Pelvic girdle (2), Lower limbs (60)

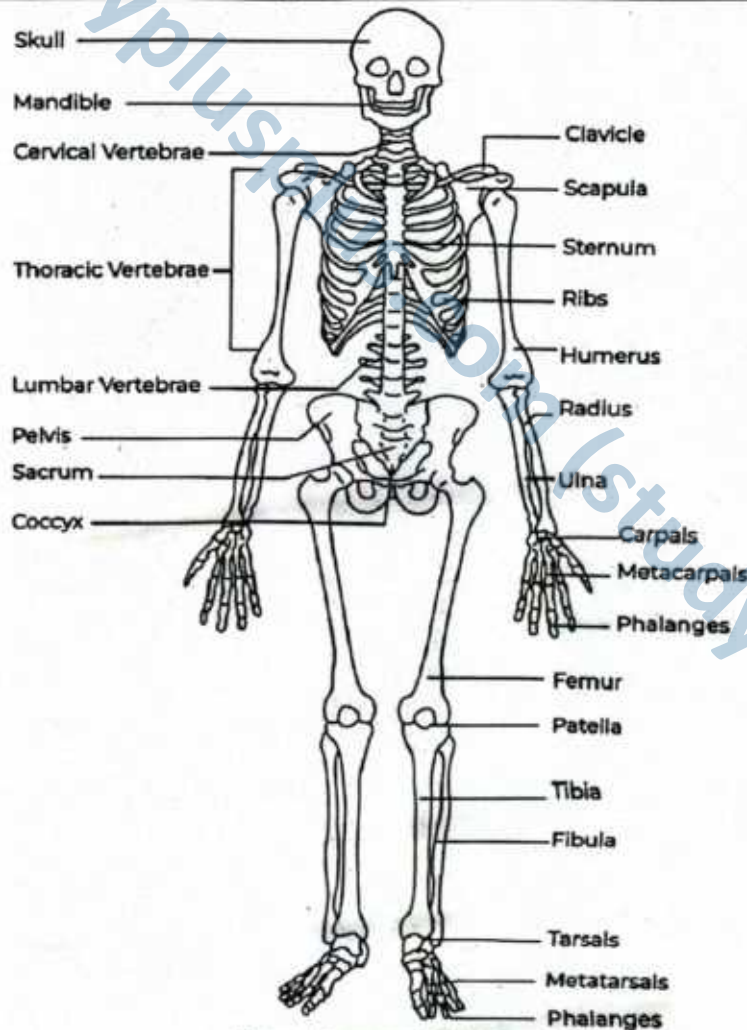


Figure: Human skeleton

Bones of Axial Skeleton

Skeletal Division	Part	Bone Names	Number of Bones
Axial Skeleton	Cranial Bones	Frontal, Parietal (×2), Temporal (×2), Occipital, Ethmoid, Sphenoid	8
	Facial Bones	Nasal (×2), Lacrimal (×2), Zygomatic (×2), Maxilla (×2), Palatine (×2), Inferior nasal concha (×2), Mandible, Vomer	14
	Ear Ossicles	Malleus (×2), Incus (×2), Stapes (×2)	6
	Hyoid Bone	Hyoid (does not articulate with any other bone)	1
	Vertebral Column	Cervical (7), Thoracic (12), Lumbar (5), Sacral (5 fused), Coccygeal (4 fused)	33 (24 + 9 fused)
	Rib Cage	12 pairs of ribs: True ribs (1st–7th), False ribs (8th–10th), Floating ribs (11th–12th); Sternum: manubrium, body, xiphoid process	25
Total Axial Skeleton			80

Bones of Appendicular Skeleton

Division	Part	Bone Names	Number of Bones
Pectoral Girdle	Shoulder Girdle	Clavicle (×2), Scapula (×2)	4
Upper Limbs	Arm	Humerus (×2)	2
	Forearm	Radius (×2), Ulna (×2)	4
	Wrist (Carpals)	Scaphoid, Lunate, Triquetrum, Pisiform, Trapezium, Trapezoid, Capitate, Hamate (×2 of each = 8 bones per hand)	16
	Palm (Metacarpals)	Metacarpal bones I–V in each hand (×5 per hand)	10
	Fingers (Phalanges)	Proximal, Middle, Distal phalanges in each finger (except thumb with 2); 14 phalanges per hand	28
Pelvic Girdle	Hip Girdle	Coxal bones (×2), each formed by fusion of three bones: Ilium, Ischium, and Pubis	2
Lower Limbs	Thigh	Femur (×2)	2
	Knee Cap	Patella (×2)	2
	Leg	Tibia (×2), Fibula (×2)	4
	Ankle (Tarsals)	Calcaneus, Talus, Navicular, Cuboid, Medial, Intermediate, and Lateral cuneiforms (×2 of each = 7 per foot)	14
	Foot (Metatarsals)	Metatarsal bones I–V in each foot (×5 per foot)	10
	Toes (Phalanges)	Proximal, Middle, Distal phalanges in each toe (except big toe with 2); 14 phalanges per foot	28
Total appendicular skeleton			126

SLO [B-12-R-64]

Describe three types of joints i.e., fibrous joints, cartilaginous joints and synovial joints and give example of each.



JOINTS

A **joint**, or **articulation**, is the site where two bones, or a bone and cartilage, meet. Joints enable movement and provide mechanical support. The scientific study of the structure and function of joints is known as **arthrology**. Joints are broadly classified into three categories based on the type of connective tissue and the degree of movement they allow:

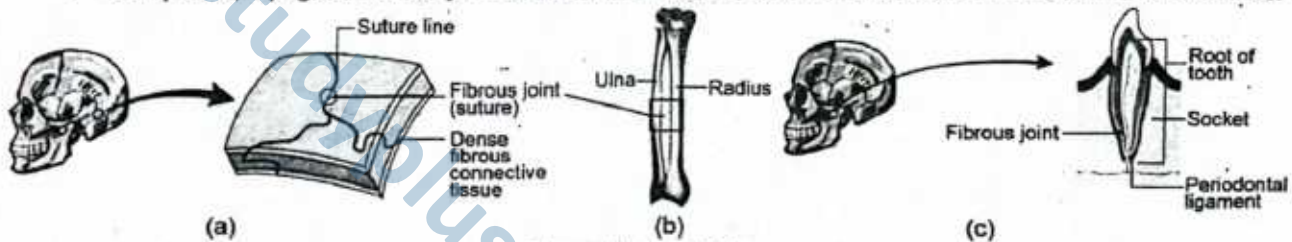
Fibrous joints

1. Fibrous Joints (Immovable Joints)

In fibrous joints, adjacent bones are directly connected by **fibrous connective tissue** primarily composed of **collagen fibers**. These joints lack a joint cavity, and the bones are either tightly fitted or slightly spaced. They allow virtually no movement.

Examples of fibrous joints:

- **Sutures** between most cranial bones of the skull.
- **Syndesmoses** between the shafts of the **radius** and **ulna** in the forearm, and **tibia** and **fibula** in the leg.
- **Gomphosis**, a peg-and-socket joint between the **root of a tooth** and its socket in the **maxilla** or **mandible**.

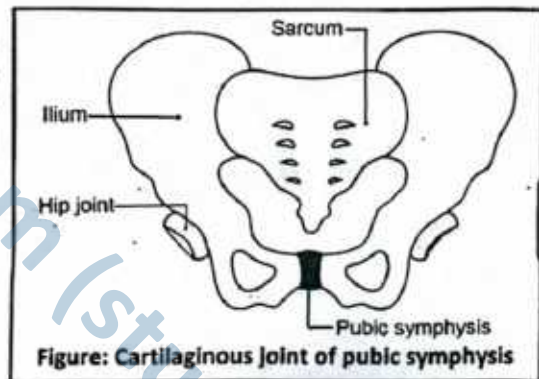


2. Cartilaginous Joints (Slightly Movable Joints)

These joints connect bones via **cartilage**, a tough yet flexible connective tissue. Like fibrous joints, cartilaginous joints lack a joint cavity. The bones are united by either **hyaline cartilage** or **fibrocartilage**, allowing limited movement.

Examples of cartilaginous joints:

- **Costal cartilages**, which connect ribs to the sternum.
- **Intervertebral discs** between the vertebrae of the spinal column.
- **Pubic symphysis**, where the two hip bones meet anteriorly.



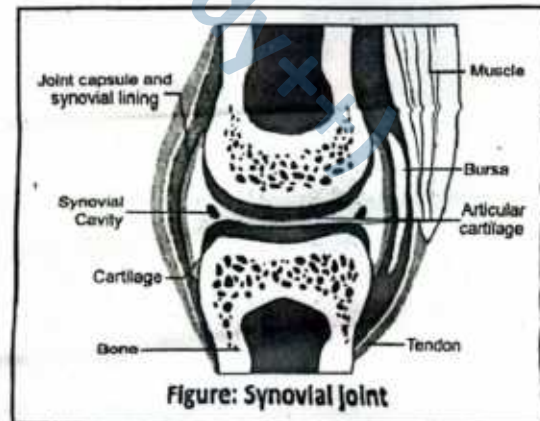
3. Synovial Joints (Freely Movable Joints)

Synovial joints permit a wide range of movements. They feature:

- **Hyaline cartilage** covering the articulating surfaces of the bones to minimize friction.
- A **joint capsule** surrounding the joint, composed of:
 - An **outer fibrous layer** reinforced by **ligaments**.
 - An **inner synovial membrane**, which secretes **synovial fluid** for lubrication and nourishment.

Examples of synovial joints:

- **Hinge joints** – allow flexion and extension (e.g., elbow, knee, fingers).
- **Ball-and-socket joints** – allow rotational and multi-directional movement (e.g., shoulder, hip).



- **Pivot joints** – permit rotation (e.g., between the atlas and axis vertebrae in the neck).
- **Gliding joints** – allow sliding movements (e.g., carpals in the wrist, tarsals in the ankle, and facet joints in the spine).

Feature	Fibrous Joints	Cartilaginous Joints	Synovial Joints
Movement Allowed	Immovable	Slightly movable	Freely movable
Joint Cavity	Absent	Absent	Present
Tissue Connecting Bones	Dense fibrous connective tissue (mainly collagen)	Cartilage (hyaline or fibrocartilage)	Hyaline cartilage + joint capsule with synovial fluid
Examples	Sutures in skull, joints between radius & ulna, gomphosis (tooth in socket)	Intervertebral discs, pubic symphysis, costal cartilage	Elbow, knee, hip, shoulder, wrist, ankle
Shock Absorption	Minimal	Moderate	Excellent (due to synovial fluid and cartilage)

SLOs Based (MCQs)

HUMAN SKELETON AND JOINTS)

- Which bones form the cranium?**
 - Frontal, Parietal, Maxilla
 - Frontal, Parietal, Temporal, Occipital, Ethmoid, Sphenoid
 - Mandible, Vomer, Maxilla
 - Scapula, Clavicle, Humerus
- Which of the following bones is unpaired in the skull?**
 - Parietal
 - Temporal
 - Frontal
 - Zygomatic
- Which bone does *not* articulate with any other bone?**
 - Hyoid
 - Maxilla
 - Mandible
 - Zygomatic
- How many bones make up the human vertebral column?**
 - 28
 - 30
 - 33
 - 36
- The first and second cervical vertebrae are called:**
 - Atlas and Axis
 - Axis and Atlas
 - Sacrum and Coccyx
 - Cervix and Cranium
- The number of thoracic vertebrae is:**
 - 5
 - 7
 - 12
 - 10
- The rib cage in humans is composed of how many pairs of ribs?**
 - 10
 - 12
 - 14
 - 7
- Which ribs are known as floating ribs?**
 - 1st and 2nd
 - 5th and 6th
 - 11th and 12th
 - 8th and 9th
- The bones of the pelvic girdle fuse to form which structure?**
 - Acetabulum
 - Coxa
 - Ilium
 - Pubis
- Which bone fits into the acetabulum of the pelvis?**
 - Tibia
 - Femur
 - Patella
 - Fibula
- The bone forming the kneecap is called:**
 - Tibia
 - Fibula
 - Patella
 - Tarsal
- How many carpal bones are present in one wrist?**
 - 5
 - 7
 - 8
 - 10
- Which bones make up the forearm?**
 - Humerus and Radius
 - Radius and Ulna
 - Scapula and Clavicle
 - Ulna and Femur
- Which bones form the pectoral girdle?**
 - Ilium and Ischium
 - Femur and Tibia
 - Clavicle and Scapula
 - Humerus and Radius
- How many bones are found in both hands together?**
 - 54
 - 27
 - 20
 - 48
- Which of the following is the longest bone in the human body?**
 - Tibia
 - Radius
 - Humerus
 - Femur
- Which vertebrae articulate with the ribs?**
 - Cervical
 - Thoracic
 - Lumbar
 - Coccygeal



18. Which bone of the forearm is on the thumb side?
A. Ulna B. Radius
C. Humerus D. Carpals
19. The human foot has how many phalanges in total?
A. 14 B. 28
C. 10 D. 20
20. The metatarsals are found in the:
A. Ankle B. Palm
C. Sole D. Wrist
21. Which type of joint allows the greatest range of movement?
A. Fibrous joint B. Cartilaginous joint
C. Synovial joint D. Suture
22. The joint between the root of a tooth and the jawbone is an example of:
A. Synovial joint B. Cartilaginous joint
C. Fibrous joint D. Hinge joint
23. Which structure secretes synovial fluid in a synovial joint?
A. Ligament B. Perichondrium
C. Synovial membrane D. Tendon
24. Cartilaginous joints are best described as:
A. Freely movable B. Fixed and rigid
C. Slightly movable D. Rotational joints
25. An example of a cartilaginous joint is:
A. Knee joint B. Elbow joint
C. Pubic symphysis D. Suture

Answer Key

1. B	2. C	3. A	4. C	5. A	6. C	7. B	8. C	9. B	10. B	11. C	12. C	13. B
14. C	15. A	16. D	17. B	18. B	19. B	20. C	21. C	22. C	23. C	24. C	25. C	

Short Questions

Q.1 What is the axial skeleton?

Ans. The axial skeleton forms the **central axis of the body**. It is made up of the **skull, vertebral column, and rib cage (including sternum)**. In total, it consists of about **80 bones**. Its main function is to **protect vital organs** such as the brain (skull), spinal cord (vertebral column), heart, and lungs (rib cage). It also provides **structural support and posture** by forming the upright framework of the body. Additionally, it serves as an **attachment site for muscles** involved in movement, breathing, and head support. By forming the body's central core, the axial skeleton ensures stability and protection while allowing limited flexibility.

MARKING SCHEME (3 Marks)

- Definition 2 marks; role 1 mark

Q.2 Name the bones of the cranium.

Ans. The cranium consists of **eight bones** that enclose and protect the brain. Four of them are **unpaired bones**: the **frontal bone, occipital bone, ethmoid bone, and sphenoid bone**. These occur singly in the skull. The remaining four are **paired bones**: two **parietal bones** and two **temporal bones**, which occur on both sides of the skull. Together, these bones form a strong protective casing around the brain, provide attachment for facial and neck muscles, and house structures such as the ear. The cranium also plays a role in determining the shape of the head and face.

MARKING SCHEME (3 Marks)

- Paired bones 2 marks; unpaired bones 1 mark

Q.3 What is the function of the vertebral column?

Ans. The vertebral column, or backbone, is made up of **33 vertebrae** arranged into cervical, thoracic, lumbar, sacral, and coccygeal regions. Its primary function is **support**—it forms the main structural framework of the body, bearing the weight of the head and trunk. Secondly, it **protects the spinal cord**, which runs through the vertebral canal, safeguarding it from injury. Thirdly, it provides **flexibility and mobility**, allowing bending, twisting, and other movements while maintaining **strength**. In addition, it serves as an **attachment site for muscles and ribs**, contributing to posture and locomotion.

MARKING SCHEME (3 Marks)

- 1 mark each for three main functions

Q.4 What is the significance of the hyoid bone?

Ans. The **hyoid bone** is a **U-shaped bone** located in the neck, just below the mandible (lower jaw) and above the larynx. It is unique because it **does not articulate with any other bone**, being suspended by muscles and ligaments. Its main role is to **support the tongue** and provide

MARKING SCHEME (3 Marks)

- Location 1 mark; role 2 marks

attachment for tongue and neck muscles. It also plays an essential role in **swallowing and speech**, as movements of the hyoid help in elevating the larynx and positioning the tongue. Because of its functional importance, any damage to the hyoid can impair speaking or swallowing.

Q.5 What is the difference between true, false, and floating ribs?

Ans. The human rib cage consists of **12 pairs of ribs**, classified into three groups. **True ribs (1st–7th pairs)** attach directly to the sternum through their own costal cartilages, providing strong protection and stability. **False ribs (8th–10th pairs)** do not connect directly; instead, their cartilages join with the cartilage of the 7th rib before reaching the sternum. **Floating ribs (11th–12th pairs)** have no anterior attachment to the sternum at all; they are only attached posteriorly to the vertebrae. True ribs mainly protect thoracic organs, while false and floating ribs provide flexibility and protect deeper structures such as the kidneys.

MARKING SCHEME (3 Marks)

- 1 mark for each type

Q.6 What bones form the pectoral girdle?

Ans. The pectoral girdle is made up of **two clavicles (collarbones)** and **two scapulae (shoulder blades)**, one on each side of the body. The clavicles are long, slender bones that connect the sternum to the scapulae, providing support for the shoulders. The scapulae are large, flat, triangular bones located at the back of the rib cage. Together, they form a strong yet flexible framework that **supports the upper limbs**, allows attachment of muscles, and enables a wide range of **shoulder and arm movements**. This mobility is essential for daily activities such as lifting, pushing, and throwing.

MARKING SCHEME (3 Marks)

- Names of the bones 3 marks

Q.7 What are carpals, metacarpals, and phalanges?

Ans. The bones of the **hand** are divided into three groups. **Carpals** are the small bones of the wrist, arranged in two rows, with **8 bones in each hand**. **Metacarpals** form the palm of the hand, consisting of **5 long bones**, one for each digit. **Phalanges** are the bones of the fingers, with **14 in each hand** (three in each finger and two in the thumb). This arrangement provides both **strength and flexibility**, allowing precise hand movements, gripping, and manipulation of objects.

MARKING SCHEME (3 Marks)

- Carpals 1 mark; metacarpals 1 mark; phalanges 1 mark

Q.8 Describe the structure of the pelvic girdle.

Ans. The **pelvic girdle** supports the weight of the body and connects the lower limbs to the axial skeleton. Each half of the girdle is formed by a **hip bone**, which is the fusion of three bones: the **ilium** (broad, upper part), the **ischium** (posterior, lower part), and the **pubis** (anterior part). The two hip bones meet at the **pubic symphysis**, a cartilaginous joint at the midline. On the outer side of each hip bone lies the **acetabulum**, a deep socket that forms the joint with the femur. This structure provides strength, supports body weight, and protects pelvic organs.

MARKING SCHEME (3 Marks)

- Ilium 1 mark; ischium 1 mark; pubis 1 mark

Q.9 Which bones make up the lower limb?

Ans. The **lower limb** consists of several major bones adapted for strength, support, and movement. The **femur** (thigh bone) is the longest and strongest bone in the body. The **patella** (kneecap) protects the knee joint. The **tibia** (shin bone) and **fibula** form the lower leg, with the tibia bearing most of the weight. The **tarsals** (7 bones) form the ankle, the **metatarsals** (5 bones) form the foot arch, and the **phalanges** (14 bones) form the toes. Together, these bones enable standing, walking, running, and balancing the body.

MARKING SCHEME (3 Marks)

- Names of the bones 3 marks

Q.10 What is the difference between tibia and fibula?

Ans. The **tibia**, commonly known as the shin bone, is the **larger and stronger bone** of the lower leg. It is the main **weight-bearing bone**, forming the **knee joint** with the femur above and the ankle joint with the talus below. In contrast, the **fibula** is a **slender bone** located laterally to the tibia. Although it does not bear significant weight, it provides **muscle attachment sites** and contributes to the stability of the ankle joint. Thus, the tibia is essential for support and movement, while the fibula assists in muscle function and balance.

MARKING SCHEME (3 Marks)

- Tibia 1.5 marks; fibula 1.5 marks



Q.11 What are fibrous joints and where are they found?

Ans. Fibrous joints are **immovable joints** where bones are held tightly together by **dense fibrous connective tissue** rich in collagen fibers. They have **no joint cavity**, which restricts movement and provides stability. These joints are mainly found in areas requiring firmness and protection. Examples include the **sutures of the skull**, which protect the brain; the **syndesmosis between the radius and ulna shafts**, allowing limited twisting; and the **gomphosis joint between teeth and their sockets** in the jaw.

MARKING SCHEME (3 Marks)

- Definition 1 mark; locations 2 marks

Q.12 How do synovial joints differ from other types of joints?

Ans. Synovial joints are the **freely movable joints** of the body. They differ from fibrous and cartilaginous joints because they have a **joint cavity** filled with **synovial fluid**, which lubricates and nourishes the joint. The articulating bone surfaces are covered with **hyaline cartilage** to reduce friction. These bones are enclosed within a **fibrous capsule lined by a synovial membrane**, ensuring smooth movement. Unlike immovable fibrous or slightly movable cartilaginous joints, synovial joints allow the **widest range of motion**, such as bending, rotation, and gliding.

MARKING SCHEME (3 Marks)

- Definition 1 mark; differences 2 marks

Q.13 Give two examples of cartilaginous joints and state their function.

Ans. Cartilaginous joints are joints where bones are joined by cartilage, allowing **slight but limited movement**. Examples include **intervertebral discs** between vertebrae and the **pubic symphysis** joining the two halves of the pelvis. Their primary function is to act as **shock absorbers**, reducing the impact of movement on bones. They also provide **flexibility** while maintaining strong connections between bones, making them important for stability in the spine and pelvis.

MARKING SCHEME (3 Marks)

- Two examples 2 marks; function 1 mark

SCIENCE, TECHNOLOGY AND SOCIETY CONNECTIONS**Techniques for Joint Replacement (Arthroplasty)**

Joint replacement, medically termed **arthroplasty**, involves replacing damaged or diseased joints with artificial components. A wide range of joints, including the hip, knee, shoulder, and elbow, can be surgically replaced. The artificial joints are typically composed of **metals** such as stainless steel or titanium alloys, combined with **modern polymers** like high-density polyethylene, silastic, or elastomer.

There are two main techniques used in joint replacement surgery:

- **Partial Joint Replacement (Hemi-replacement Technique):** In this procedure, the damaged articular surface of the bone on **only one side of the joint** is removed and replaced with an artificial component.
- **Total Joint Replacement Technique:** In this method, the **articular surfaces on both sides of the joint** are removed and replaced with prosthetic components to restore full function.

The prosthetic components are typically affixed to the bone using a **synthetic adhesive**, most commonly **methyl methacrylate**, to ensure secure attachment.

Relationship Between Bipedal Posture and the Human Skeleton and Musculature

The bipedal posture of humans is supported by specific adaptations in both the skeletal and muscular systems:

- **Vertebral Column Curvatures:** The natural curvatures of the vertebral column contribute to body balance in the upright stance and help distribute body weight evenly.
- **Intervertebral Discs:** These fibrocartilaginous structures provide flexibility to the spine and serve as shock absorbers during bipedal locomotion.
- **Pelvic Structure:** The shape and orientation of the pelvis, especially its articulation with the vertebral column, allow for an upright posture and efficient weight transfer during walking.

Postural Muscles: Certain muscles, especially those along the spine and lower limbs, act as **anti-gravity muscles**, continuously contracting to oppose gravitational forces. These include strong, complex **erector spinae muscles** and **gluteal muscles**, which maintain body alignment and stability.



COMMON DISORDERS OF THE HUMAN SKELETON

The human skeletal system, though highly resilient, is susceptible to a variety of disorders that impair movement, posture, and overall health. These disorders may arise from **aging, injury, poor posture, degenerative changes, nutritional deficiencies, or autoimmune conditions**. Important disorders are discussed below:

1. Slipped Disc (Herniated Disc)

Each intervertebral disc is a cushion-like structure located between adjacent vertebrae. It is composed of two distinct parts:

- **Nucleus pulposus:** a soft, gelatinous central region that provides elasticity and compressibility.
- **Annulus fibrosus:** a tough outer ring of fibrocartilage that firmly anchors the vertebrae together.

These discs act as shock absorbers during movement. A **slipped disc** (more accurately called a **herniated disc**) occurs when the annulus fibrosus ruptures, allowing the nucleus pulposus to protrude outward. This herniation may compress the spinal cord or nearby spinal nerves, leading to **severe back pain, nerve irritation, numbness, or even loss of motor function**.

Note: The term "slipped disc" is misleading, since the disc itself does not slip out of place; only its inner contents bulge abnormally.

2. Spondylosis

Spondylosis is a degenerative disorder of the vertebral column that results in stiffness, pain, and reduced mobility. It is associated with the **gradual wear and tear of intervertebral discs, ligaments, and vertebral joints**. In advanced stages, it may lead to immobility and even fusion of vertebrae.

- **Cervical spondylosis** is a common type, involving degeneration of cervical vertebrae. It often results in neck pain, stiffness, headaches, and sometimes compression of spinal nerves.
- **Risk factors** include aging, repetitive strain, and long-term poor posture.

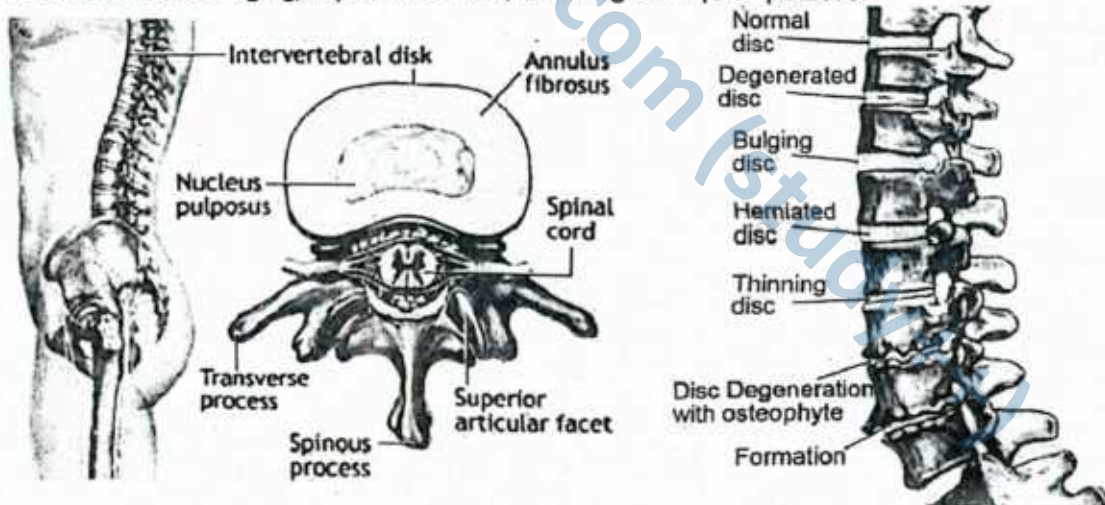


Figure: Structure of (a) Intervertebral disc (b) slipped disc

3. Sciatica

Sciatica refers to pain and discomfort caused by pressure on the **sciatic nerve**, the longest nerve in the human body. The pain typically radiates from the lower back through the buttocks and down the legs, and may be accompanied by numbness, tingling, or muscle weakness.

- **Causes:** Most commonly a herniated disc, but also spinal stenosis, bone spurs, pelvic fractures, or tumors pressing on the nerve.
- **Symptoms:** Sharp, shooting pain down the leg, sometimes so severe that normal walking or sitting becomes difficult.

4. Arthritis

Arthritis is a general term for **inflammation of joints**, leading to pain, stiffness, swelling, and difficulty in mobility. It can significantly impair everyday activities such as walking, climbing stairs, or standing up from a seated position.

Major Types Include:

- **Osteoarthritis:** The most common type, caused by progressive degeneration of articular cartilage. It typically affects weight-bearing joints such as the knees, hips, and spine, producing pain after movement and a "grating" sound in joints.
- **Rheumatoid arthritis:** An **autoimmune disease** in which the immune system attacks the synovial membrane, causing swelling, deformity, and chronic pain.
- **Gouty arthritis:** Caused by excessive uric acid in the blood. Crystals of sodium urate deposit in joints, especially the big toe, leading to sudden episodes of severe pain, redness, and swelling.

5. Osteoporosis

Osteoporosis is a skeletal disorder characterized by **low bone mass, increased bone fragility, and a high risk of fractures**. Bones become porous and brittle, particularly in the spine, hip, and wrist.

- **Why it occurs:**
 - Bone mass normally peaks between ages **20–25 years**.
 - With advancing age—especially after menopause in women—bone resorption exceeds bone deposition.
 - This imbalance gradually weakens the bones.
- **Risk factors:** Aging, deficiency of calcium and vitamin D, hormonal changes (low estrogen in women), sedentary lifestyle, and prolonged use of corticosteroid medications.
- **Prevention:**
 - Ensuring adequate **calcium and vitamin D intake** during childhood and adolescence, when bone mass is being built.
 - Engaging in **weight-bearing exercises** to strengthen bones.
 - Attaining a **high peak bone mass** in early adulthood helps tolerate natural age-related bone loss later in life.

The major disorders of the skeletal system—**slipped disc, spondylosis, sciatica, arthritis, and osteoporosis**—demonstrate the importance of maintaining skeletal health through **proper nutrition, posture, exercise, and preventive care**. Early diagnosis and treatment can reduce the severity of these conditions and improve quality of life.

SLO [B-12-R-66]

State different types of fractures (simple, compound and complicated) and describe the repair process of simple fractures.

BONE FRACTURES

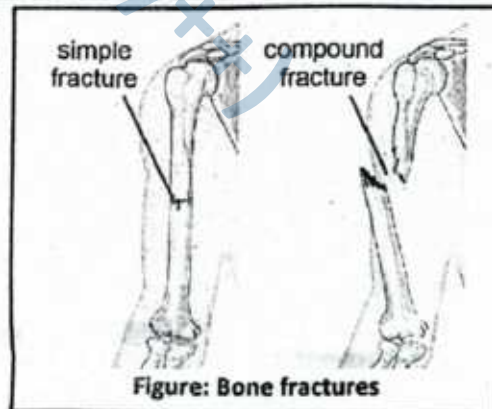
A fracture is the medical term for a broken bone. They occur when the physical force exerted on the bone is stronger than the bone itself. So bones break when they cannot withstand a force or trauma applied to them.

Common types of fractures

A **fracture** is a break in bone integrity caused when the applied force exceeds the bone's mechanical strength.

Common Types Include:

- **Simple (closed) fracture** – The bone is broken, but the skin remains intact.
- **Compound (open) fracture** – Bone ends pierce the skin, increasing the risk of infection.
- **Complicated fracture** – A fracture that damages surrounding tissues or organs.



Bone Repair

Bone is a dynamic, living tissue capable of self-repair following injury. The process of bone repair after a fracture involves a highly coordinated sequence of biological events that restore the bone's original structure and function. The healing process typically proceeds through four overlapping stages:

1. Haematoma Formation (Clot Formation)

- Immediately after a bone fractures, blood vessels within the bone and surrounding periosteum are ruptured, resulting in hemorrhage at the fracture site.
- This leads to the formation of a **haematoma**, a localized mass of clotted blood. The haematoma causes swelling, inflammation, and cellular death in the area due to loss of blood supply.

2. Fibrocartilaginous Callus Formation (Soft Callus Formation)

- Within a few days, capillaries grow into the haematoma and phagocytic cells begin clearing dead cells and debris.
- Fibroblasts and chondroblasts invade the area and produce collagen fibers and fibrocartilage.
- Osteoblasts begin to proliferate from the periosteum and endosteum, forming a **fibrocartilaginous (soft) callus** that temporarily stabilizes the fracture. This phase lasts about **3–4 weeks**.

3. Bony Callus Formation (Hard Callus/Ossification)

- As healing progresses, the soft callus is gradually replaced by a **bony (hard) callus** composed of spongy bone.
- Osteoblasts convert the cartilage into woven bone through **endochondral ossification**. This hard callus formation strengthens the fracture site and typically spans **6–8 weeks**, depending on age, bone type, and overall health.

4. Bone Remodelling

- In the final phase, the bony callus is remodeled into compact bone.
- Osteoclasts resorb excess bone tissue, and osteoblasts lay down new lamellar bone, restoring the bone's original shape and internal structure.
- The remodeling phase may take **several months** and ensures that the repaired bone can withstand normal mechanical stresses.

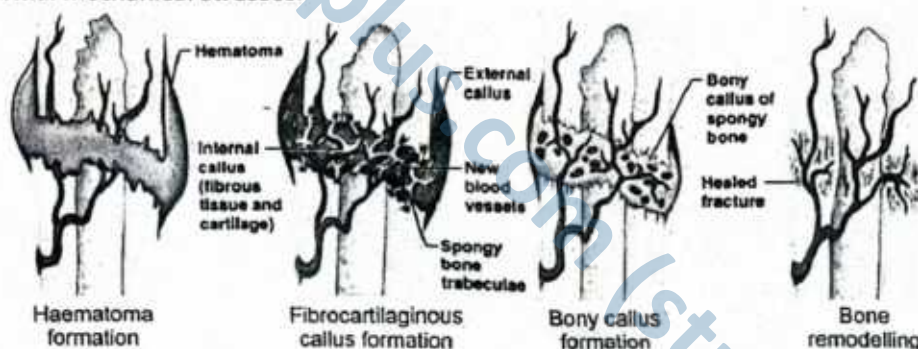


Figure: Bone repair

SCIENCE, TECHNOLOGY AND SOCIETY CONNECTIONS

Relate improper posture to bone/joint problems

Improper posture causes increased stress on joints and their supporting structures resulting in injury, pain and early degeneration of bones and joints.

SLO [B-12-R-67]

Describe the injuries in joints (dislocation and sprain) and their first aid treatment.

INJURIES IN JOINTS

Joints, being the points of articulation between bones, are particularly vulnerable to injury. **Torsion (twisting forces)** or a **sudden impact from the side** of a joint can severely compromise its structural integrity. Two of the most common joint injuries are **dislocation** and **sprain**, both of which differ in mechanism and severity but can cause significant pain and functional limitation.



1. Dislocation of Joints

A **dislocation** occurs when the bones forming a joint are forcibly displaced from their normal alignment. This injury usually results from sudden trauma, such as a fall, sports accident, or violent twist, that overwhelms the stabilizing structures of the joint. In severe cases, dislocation may be accompanied by tearing of the **ligaments, tendons, and muscles** surrounding the joint.

- **Causes:**
 - Direct trauma (falls, road accidents, or sports injuries).
 - Congenital weakness of ligaments or shallow joint cavities.
 - Underlying conditions such as **rheumatoid arthritis**, which weaken joint stability.
- **Symptoms:**
 - Intense pain and tenderness.
 - Visible deformity or abnormal bone positioning.
 - Swelling around the joint.
 - Limited or complete loss of movement.
- **Treatment:**
 - Dislocations must be reduced (manually repositioned) only by trained medical personnel. Improper handling may worsen tissue damage or injure nerves and blood vessels. In some cases, surgical intervention is required to repair or tighten overstretched ligaments. Rehabilitation is often necessary to restore mobility and strength.

2. Sprain

A **sprain** is an injury involving the **stretching or tearing of ligaments**, the fibrous tissues that connect bones within a joint. Unlike dislocation, the bones remain in their proper alignment, but the ligaments become damaged due to excessive force or sudden twisting.

- **Commonly Affected Joints:**
 - **Ankle** (most frequent in sports or missteps).
 - **Knee** (often during sudden pivots or impact).
 - **Wrist** (common in falls on an outstretched hand).
- **Symptoms:**
 - Pain and tenderness.
 - Swelling and bruising around the joint.
 - Difficulty in moving or bearing weight on the affected limb.
- **Treatment:**

Most sprains are treated conservatively using the **R.I.C.E. protocol**:

 - **Rest** the joint to prevent further damage.
 - **Ice** application to reduce swelling and pain.
 - **Compression** with an elastic bandage or brace for support.
 - **Elevation** of the injured limb above heart level to minimize swelling.

Severe sprains, where ligaments are completely torn, may require prolonged immobilization or surgical repair, followed by physiotherapy.

3. First Aid Treatment for Skeletal Injuries

Prompt and appropriate **first aid** is crucial for improving recovery outcomes in skeletal injuries, including dislocations, sprains, and fractures.

First Aid Guidelines:

- **Immobilize** the injured joint or limb using a sling, splint, or bandage. Do **not** attempt to realign a dislocated joint or fractured bone.
- Apply **ice packs** intermittently to reduce pain, swelling, and inflammation.
- Keep the injured person in a **comfortable position** and provide reassurance to minimize anxiety and shock.
- Observe for signs of **circulatory or nerve damage**, such as numbness, pallor, or loss of sensation below the injury.
- For **serious injuries** such as compound fractures or dislocations of **weight-bearing joints** (hip, ankle, or knee), seek immediate **emergency medical assistance** for safe transport and treatment.



1. Which cells are primarily responsible for bone repair?
A. Osteocytes B. Osteoclasts
C. Osteoblasts D. Chondrocytes
2. The first step in bone repair involves:
A. Callus formation B. Bone remodeling
C. Hematoma formation D. Cartilage deposition
3. Which structure helps stabilize joints and may be torn during dislocation?
A. Tendon B. Ligament
C. Periosteum D. Synovial membrane
4. A dislocated joint results when:
A. A bone breaks B. A tendon tears
C. Bones move out of alignment at a joint
D. Cartilage degenerates
5. A sprain is an injury to the:
A. Muscle B. Bone
C. Ligament D. Tendon
6. Which of the following is not a symptom of joint dislocation?
A. Swelling B. Immobility
C. Red rash D. Intense pain
7. Rheumatoid arthritis may lead to:
A. Bone fracture B. Muscle paralysis
C. Joint dislocation D. Tendon rupture
8. The correct order of bone repair is:
A. Callus formation → Hematoma → Remodeling
B. Hematoma → Callus formation → Bone remodeling
C. Inflammation → Cartilage growth → Repair
D. Bone remodeling → Hematoma → Fibrosis
9. Immobilization of a dislocated joint should be:
A. Attempted by a friend
B. Done after manual reduction
C. Done without manipulation
D. Avoided entirely
10. What is the function of an ice pack in first aid for joint injuries?
A. Promote blood flow
B. Prevent infection
C. Reduce swelling and pain
D. Increase joint mobility
11. Which joint injury often requires surgery to repair torn ligaments?
A. Minor sprain B. Bone bruise
C. Severe dislocation D. Synovial inflammation
12. Which term refers to the manual return of a dislocated bone to its position?
A. Dislocation B. Realignment
C. Reduction D. Rotation
13. A compound fracture:
A. Is a small crack in the bone
B. Involves dislocation
C. Breaks the skin surface
D. Only affects cartilage
14. During which phase of bone healing does fibrocartilage form?
A. Hematoma B. Callus formation
C. Remodeling D. Ossification
15. What is the main role of ligaments in joints?
A. Connect bone to muscle B. Produce synovial fluid
C. Connect bone to bone D. Cushion the joint
16. Which first aid method provides support to an injured joint?
A. Ice application B. Sling or splint
C. Painkiller D. Massage
17. Which joint disorder results from overstretching of ligaments without bone displacement?
A. Arthritis B. Sprain
C. Dislocation D. Fracture
18. Which of the following is not a cause of joint dislocation?
A. Trauma B. Congenital defect
C. Tendon inflammation D. Rheumatoid arthritis
19. After dislocation, the joint becomes:
A. More rigid B. Immobile and painful
C. Easily stretched D. Stronger
20. The long-term phase of bone healing involves:
A. Reduction B. Remodeling
C. Fracture D. Hematoma

Answer Key

1. C	2. C	3. B	4. C	5. C	6. C	7. C	8. B	9. C	10. C
11. C	12. C	13. C	14. B	15. C	16. B	17. B	18. C	19. B	20. B



Q.1 What is the correct sequence of events during bone healing after a fracture?

Ans. Bone healing occurs in three main stages. First, a **hematoma** forms at the fracture site due to ruptured blood vessels, bringing inflammatory cells to start repair. Next, a **fibrocartilaginous callus** develops, bridging the broken bone ends with cartilage and collagen. Finally, the callus is replaced by a **bony callus**, and the bone undergoes **remodeling**, where compact bone restores the original shape and strength.

MARKING SCHEME (3 Marks)

- Names of all steps with brief description

Q.2 How does a dislocation occur and what are its symptoms?

Ans. A **dislocation** occurs when bones in a joint are pushed out of their normal position, usually from **trauma, falls, or sudden force**. The main symptoms include **intense pain, swelling and immobility**, and often a **visible deformity** of the affected joint. Immediate immobilization and medical treatment are essential to avoid nerve or blood vessel damage.

MARKING SCHEME (3 Marks)

- Cause 1 mark; symptoms 2 marks

Q.3 What is the difference between a sprain and a fracture?

Ans. A **sprain** is an injury to ligaments caused by excessive stretching or tearing, without displacement of bones. It leads to **pain, swelling, and reduced joint mobility**. In contrast, a **fracture** is an actual **break in the bone**, which may be simple (closed) or compound (open, piercing the skin). Unlike sprains, fractures require longer healing and immobilization of the bone.

MARKING SCHEME (3 Marks)

- Definition of sprain 1.5 marks;
definition of fracture 1.5 marks

SLO [B-12-R-68]

Compare smooth muscles, cardiac muscles and skeletal muscles.

MUSCLES

Muscles are specialized contractile tissues that enable movement of body parts or the entire body. They are also responsible for maintaining posture, moving body fluids, and generating heat. The scientific study of muscles is known as **myology**. Muscle tissue is classified into three main types: **smooth muscle, cardiac muscle, and skeletal muscle**.

Types of Muscles

There are three types of muscle tissues: smooth, cardiac and skeletal.

► Smooth Muscles

- Smooth muscles are found in the walls of internal organs such as the digestive tract, reproductive organs, urinary tract, and blood vessels.
- These muscles are **involuntary** and not under conscious control.
- Each smooth muscle cell is **spindle-shaped**, with a **single, centrally located nucleus**.
- Myofilaments in smooth muscle are **not arranged into sarcomeres**, hence they appear **non-striated** under the microscope.
- These cells also contain **non-contractile intermediate filaments**, contributing to their structural support.

► Cardiac Muscles

- Found **exclusively in the heart**, cardiac muscles are **involuntary** and function to pump blood throughout the body.
- They are **striated**, like skeletal muscles, due to the organized arrangement of myofilaments into sarcomeres.
- Cardiac muscle cells are **branched**, usually with a **single nucleus** near the center.
- Adjacent cardiac cells are connected by **intercalated discs**, which contain **gap junctions**. These allow action potentials to pass rapidly from cell to cell, enabling coordinated contractions.

► Skeletal Muscles

- Skeletal muscles are attached to bones and are responsible for **voluntary movements and locomotion**.



- They are composed of long, cylindrical, multinucleated cells called **muscle fibres**.
- These fibres exhibit a characteristic **striated appearance** due to the alternating light (I bands) and dark (A bands) regions.
- Each muscle fibre is enclosed by a plasma membrane called the **sarcolemma**, and contains cytoplasm known as **sarcoplasm**, which houses the **sarcoplasmic reticulum** and **myofibrils**.



Figure: Types of muscles

Comparison of Three Types of Muscle Tissues

Property	Smooth Muscles	Cardiac Muscles	Skeletal Muscles
Muscle appearance	Unstriated	Irregular striped	Regular striped
Cell shape	Spindle	Branched	Spindle or cylindrical
Number of nuclei	One per cell	One per cell	Many per cell
Speed of contraction	Slow	Intermediate	Slow to rapid
Contraction caused by	Nervous system	Spontaneous	Nervous system
Function	Controls movement of substances through hollow organs	Pumps blood	Move skeleton
Voluntary control	Usually no control	Usually no control	Have control

SLO [B-12-R-69]

Annotate the ultrastructure of the skeletal muscle.

ULTRASTRUCTURE OF SKELETAL MUSCLES

Skeletal muscles are the organs of voluntary movement. They are highly organized structures, both at the macroscopic and microscopic levels, enabling them to generate precise and powerful contractions.

Externally, a skeletal muscle is wrapped in a connective tissue sheath called the **epimysium**. Each muscle is composed of multiple bundles known as **fascicles**, each fascicle being surrounded by **perimysium**. Within a fascicle, individual **muscle fibers (muscle cells)** are elongated, cylindrical, multinucleated cells. Every muscle fiber is encased in a thin connective tissue covering called the **endomysium**. This hierarchical arrangement provides both structural strength and efficient force transmission during contraction.

Ultrastructure of Skeletal Muscle

1. Muscle Fiber and Sarcolemma

Each muscle fiber is bounded by the **sarcolemma**, a specialized plasma membrane that not only encloses the cell but also plays a crucial role in impulse conduction. The sarcolemma folds inward to form **transverse tubules (T-tubules)**, which carry electrical impulses deep into the cell, ensuring synchronized contraction of the entire fiber. The cytoplasm, known as **sarcoplasm**, contains abundant mitochondria, glycogen, and the oxygen-binding pigment **myoglobin**, all of which support the high energy demands of contraction.

2. Myofibrils and Sarcomeres

The bulk of the sarcoplasm is occupied by **myofibrils**, which are long, thread-like contractile elements running the length of the muscle fiber. Myofibrils are composed of even smaller protein filaments, termed **myofilaments**, of two main types:

- **Thin filaments (~7–8 nm):** Composed of **actin**, **tropomyosin**, and **troponin**.
- **Thick filaments (~16 nm):** Composed primarily of **myosin**.

Myofibrils are organized into repeating units called **sarcomeres**, the fundamental contractile units of skeletal and cardiac muscle. Each sarcomere is bounded by **Z-lines**. The specific arrangement of thin and thick filaments produces the characteristic **striations** seen in skeletal muscle:



- **I band:** Contains only thin filaments (appears light).
- **A band:** Spans the entire length of thick filaments, including overlap with thin filaments (appears dark).
- **H zone:** Central lighter region within the A band where only thick filaments are present.
- **M line:** Found in the middle of the H zone; contains proteins that stabilize thick filaments.

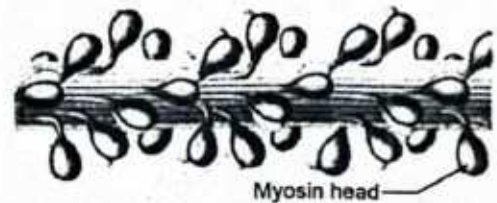
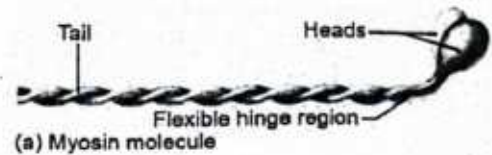
This structural arrangement allows for the sliding filament mechanism, where thin filaments slide past thick filaments to shorten the sarcomere and generate contraction.

Molecular Components of Myofilaments

1. Thick Filaments (Myosin)

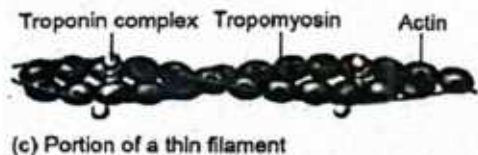
Each **myosin molecule** consists of six polypeptide chains, organized into a long tail and two protruding **globular heads**.

- The **heads** project outward, forming cross-bridges with actin during contraction.
- Each head possesses **ATPase activity**, enabling hydrolysis of ATP to power contraction.
- A thick filament contains about **300 myosin molecules**, with their tails aligned to form the shaft and heads projecting outward at opposite ends, a design crucial for bidirectional interaction with actin.



Thin Filaments (Actin, Tropomyosin, and Troponin)

- **Actin:** Exists initially as **G-actin (globular actin)**, each subunit bearing a myosin-binding site. These monomers polymerize into **F-actin (filamentous actin)** chains, which intertwine to form the helical backbone of the thin filament.
- **Tropomyosin:** A rod-shaped protein that wraps around actin filaments, covering the myosin-binding sites when the muscle is at rest.
- **Troponin:** A regulatory protein complex with three subunits:
 - **Troponin I (TnI):** Binds to actin, inhibiting contraction.
 - **Troponin T (TnT):** Binds to tropomyosin, anchoring the complex.
 - **Troponin C (TnC):** Binds calcium ions, initiating conformational changes that expose the binding sites for myosin.



This finely tuned arrangement ensures that contraction occurs only upon stimulation, preventing unnecessary energy expenditure.

2. Sarcoplasmic Reticulum and Triad System

Surrounding each myofibril is a network of **sarcoplasmic reticulum (SR)**, a specialized organelle that stores and releases calcium ions. At the junction of A and I bands, the SR forms enlarged sacs called **terminal cisternae**, which flank the **T-tubules**. Together, two terminal cisternae and one T-tubule form a **triad**, a structure critical for coupling electrical excitation with calcium release, thereby initiating contraction.

Functional Significance

The ultrastructure of skeletal muscle reflects its specialization for rapid, powerful, and voluntary contractions. The hierarchical arrangement—from connective tissue coverings to sarcomeres and molecular proteins—ensures strength, efficiency, and precise regulation. The sliding filament mechanism, supported by ATP hydrolysis and calcium regulation, makes skeletal muscle uniquely suited to perform tasks ranging from fine motor activities to vigorous exertion.

Externally muscle is covered in a connective tissue wrapping called epimysium. Each skeletal muscle consists of hundreds to thousands of muscle fibres (muscle cells). Each muscle is divided into discrete bundles of muscle cells called fascicles. The fascicle is surrounded by perimysium. Each fibre within the fascicle is covered by a layer of connective tissue called the endomysium.

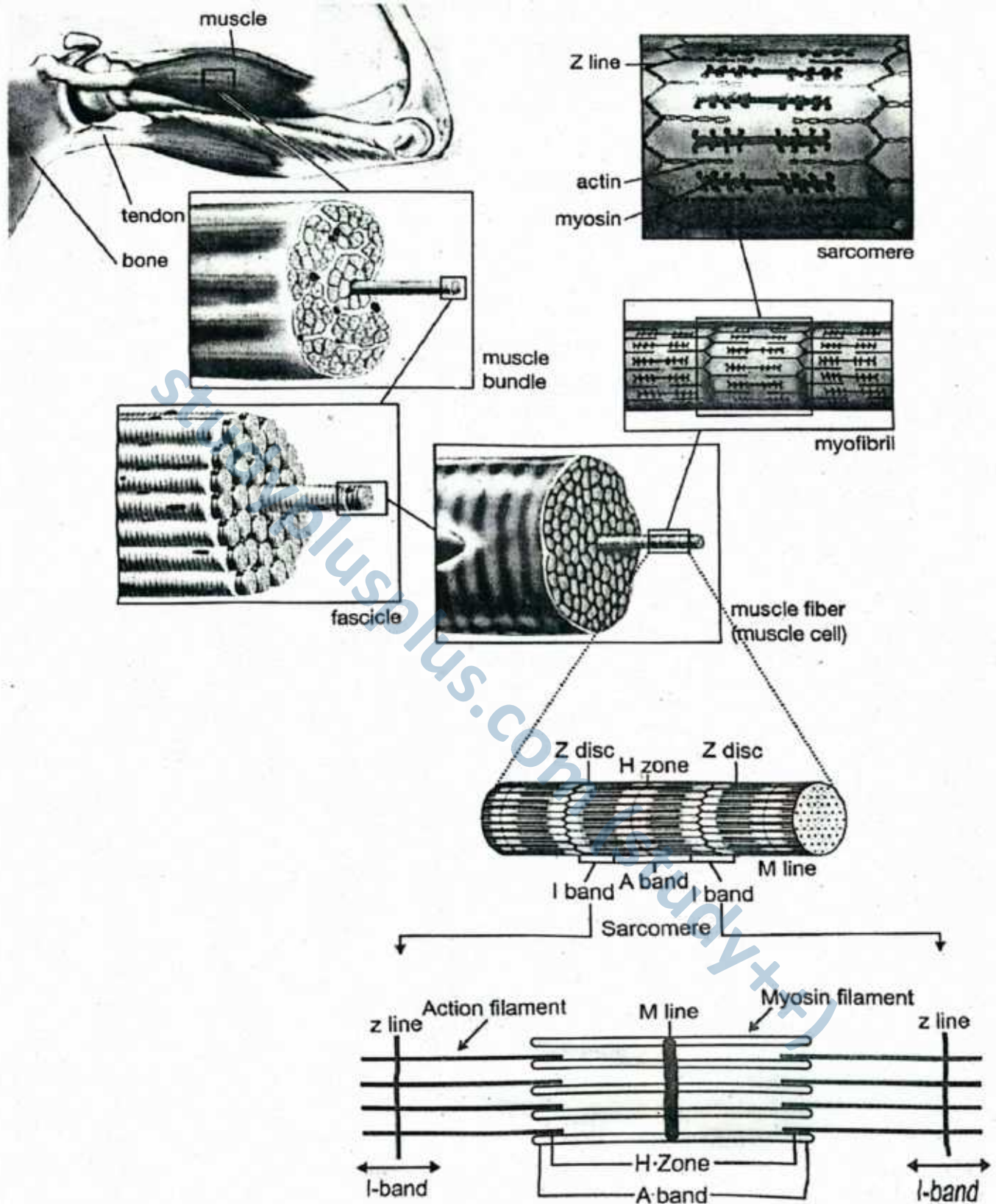
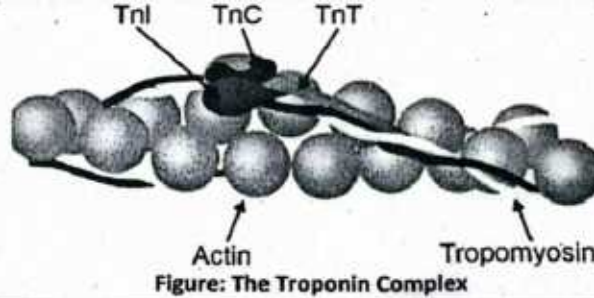


Figure: Structure of Skeletal Muscle



SLO [B-12-R-70]

Explain the sliding filaments model of muscle contraction.

MUSCLE CONTRACTION – SLIDING FILAMENT MODEL

The **sliding filament model** is the widely accepted explanation of how skeletal muscle contracts. It was independently proposed in the 1950s by **Hugh Huxley** and **Jean Hanson**, and it remains fundamental to modern muscle physiology. According to this model, **muscle contraction does not involve shortening of individual filaments**, but rather a **sliding of thin filaments (actin) past thick filaments (myosin)** within the sarcomere. This sliding action shortens the sarcomere and thereby the entire muscle fiber, producing contraction.

Sequence of Events in the Sliding Filament Model

1. **Resting State**
 - In a relaxed muscle, thin and thick filaments overlap only partially.
 - Tropomyosin covers the myosin-binding sites on actin, preventing cross-bridge formation.
2. **Stimulation and Calcium Release**
 - A **nerve impulse** triggers the release of the neurotransmitter **acetylcholine** at the neuromuscular junction.
 - This generates an **action potential** that travels along the sarcolemma and into the muscle fiber via **T-tubules**.
 - The action potential stimulates the **sarcoplasmic reticulum (SR)** to release **calcium ions (Ca^{2+})** into the sarcoplasm.
3. **Binding of Calcium to Troponin**
 - Calcium ions bind to **troponin C (TnC)** on the thin filaments.
 - This induces a conformational change in the troponin complex, which shifts **tropomyosin** away from the active sites on actin.
 - The myosin-binding sites on actin are now exposed.
4. **Cross-Bridge Formation**
 - Energized **myosin heads** (containing ADP and inorganic phosphate) attach to the exposed sites on actin, forming **cross-bridges**.
5. **Power Stroke**
 - The myosin head pivots, pulling the thin filament **toward the centre of the sarcomere (M line)**.
 - This action is powered by the energy released from **ATP hydrolysis**.
 - ADP and phosphate are released from the myosin head at the end of the stroke.
6. **Detachment of Cross-Bridge**
 - A new molecule of **ATP binds** to the myosin head.
 - This causes the cross-bridge to detach from actin.



7. Reactivation of Myosin Head

- ATP is hydrolyzed into ADP and phosphate, re-cocking the myosin head into its high-energy position.
- If calcium remains available, the cycle repeats, leading to continued contraction.

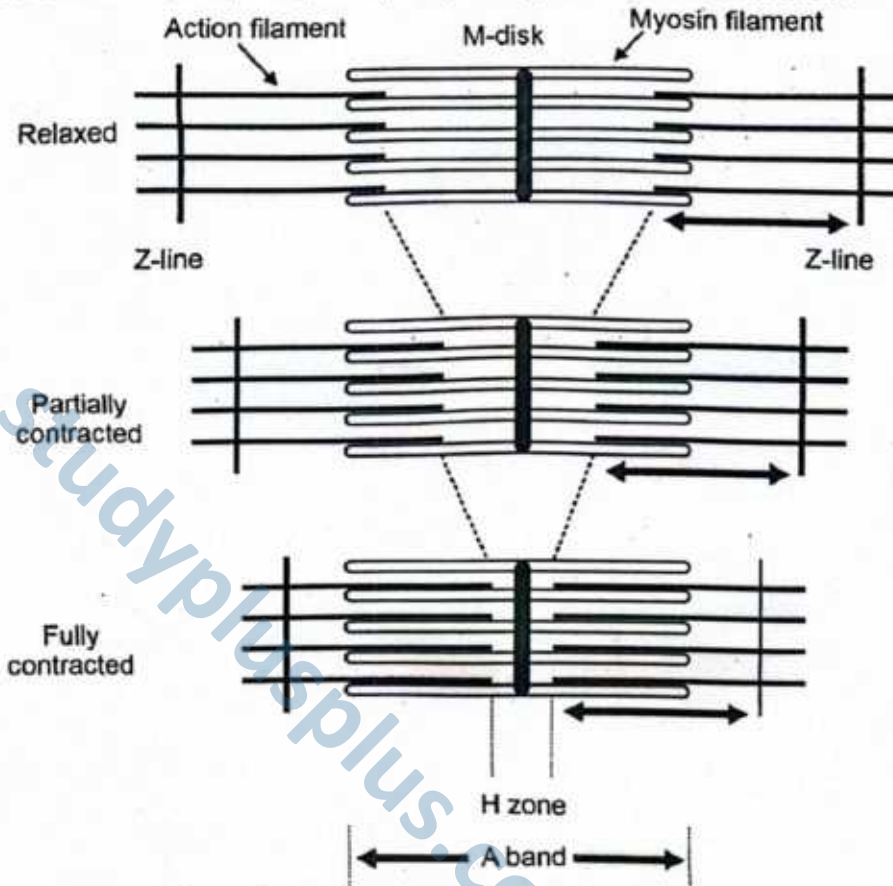


Figure: Sliding filament model of muscle contraction

Structural Changes during Contraction

As thin filaments slide inward between thick filaments:

- Sarcomere length shortens overall.
- I bands (actin only) shorten.
- H zone (myosin only) narrows or may disappear completely.
- A band (entire length of thick filaments) remains unchanged.
- Z lines move closer together, compressing the sarcomere.

Thus, contraction results not from a change in filament length but from filament sliding.

Relaxation Phase

- When neural stimulation ceases, calcium ions are actively pumped back into the sarcoplasmic reticulum.
- Troponin and tropomyosin return to their resting positions, *again covering the myosin-binding sites on actin*.
- Without cross-bridge interaction, the sarcomere lengthens back to its resting state, either passively (due to antagonistic muscles) or by elastic recoil.

Functional Significance

The sliding filament model illustrates how microscopic molecular interactions translate into macroscopic muscle movement. The cycle of **cross-bridge formation**, **power stroke**, **detachment**, and **reactivation** allows for repetitive, forceful contractions essential for locomotion, posture, and voluntary activities.

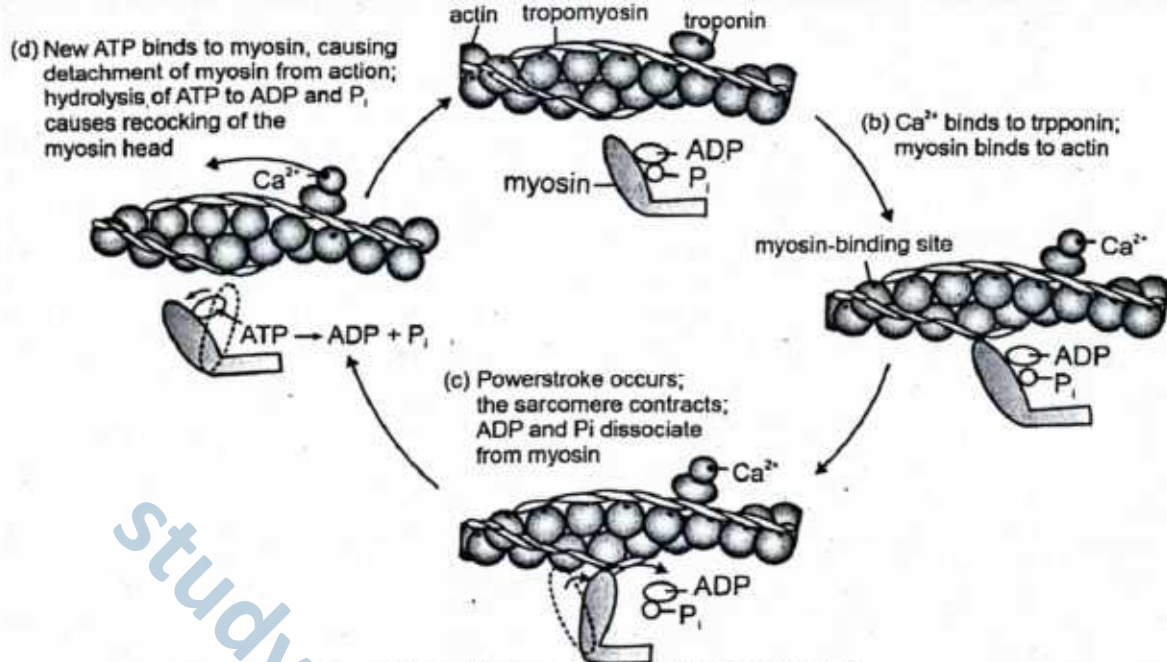


Figure: Sliding filament model of muscle contraction

SLO [B-12-R-71]

Describe the action of antagonistic muscles in the movement of knee joint.

ANTAGONISTIC ARRANGEMENT OF SKELETAL MUSCLES

Bones are joined together by strong connective tissues called **ligaments**, which stabilize the joints. Skeletal muscles are attached to bones at two points: the **origin** and the **insertion**. The **origin** is the fixed attachment point, while the **insertion** is the movable attachment, which is drawn toward the origin when the muscle contracts. The thick, central part of the muscle between origin and insertion is known as the **belly**, and it is this region that contracts during movement.

Muscles act on bones to produce movement by crossing joints. When a **flexor muscle** contracts, it bends the joint (e.g., bending the elbow or knee), whereas an **extensor muscle** straightens the joint. These muscles typically work in **antagonistic pairs**, where the contraction of one muscle (e.g., flexor) is accompanied by the relaxation of its counterpart (e.g., extensor), allowing controlled, bidirectional movement.

Movement in knee joint

The **knee joint**, or **tibiofemoral joint**, is a complex hinge joint located between the **femur** (thigh bone) and **tibia** (shin bone). It allows **flexion** and **extension**, as well as limited **rolling** and **gliding** movements.

- **Flexion** of the knee is primarily performed by the **hamstring muscles**, located at the back of the thigh. The major hamstring muscle, **biceps femoris**, originates from both the **pelvic girdle** and the **femur**. Its tendon divides and attaches to the **upper parts of the tibia and fibula**.
- **Extension** of the knee is carried out by the **quadriceps femoris**, a group of extensor muscles situated at the front of the thigh. These muscles originate from the **ilium** and **femur**, converge around the **patella** (kneecap), and insert into the **tibia**. They are crucial for walking, running, standing, and maintaining posture.

Other notable muscles involved in knee movement include:

- **Semitendinosus**: a fusiform muscle in the posterior thigh compartment.
- **Semimembranosus**: another hamstring muscle in the same region.

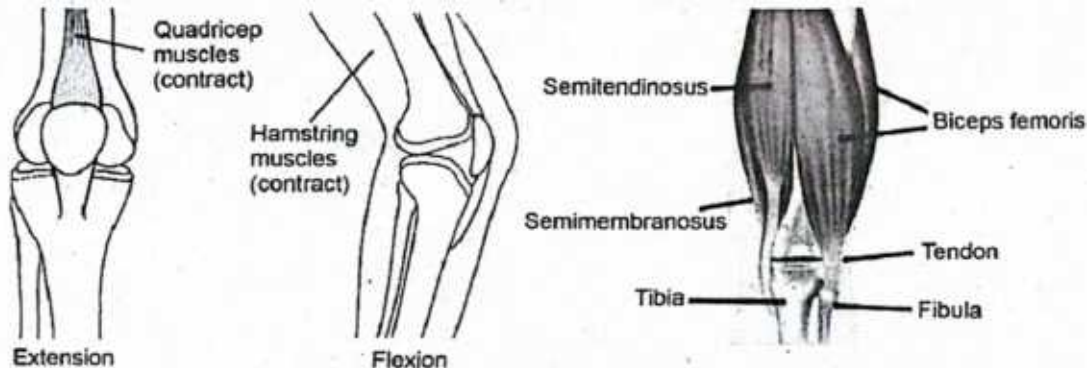


Figure: Movement at knee joint

SLO [B-12-R-72]

Explain muscle fatigue, cramps and tetany.

MUSCLE FATIGUE, CRAMPS, AND TETANY

Skeletal muscles are capable of powerful and sustained contractions, but they are also subject to limitations when energy demand exceeds supply or when physiological imbalances develop. Three important conditions associated with muscle performance are **fatigue, cramps, and tetany**.

1. Muscle Fatigue

Definition:

Muscle fatigue is the **decline in the ability of a muscle to generate force** after prolonged or intense activity, even though stimulation from the nervous system continues.

Causes and Mechanism:

- **Energy depletion:** Continuous contraction consumes large amounts of ATP. When ATP resynthesis cannot keep pace, contractile efficiency declines.
- **Lactic acid accumulation:** During anaerobic respiration, glucose is broken down into lactic acid, lowering the pH of muscle fibers. This acidic environment interferes with enzyme function and cross-bridge cycling.
- **Ion imbalance:** Prolonged activity disrupts calcium (Ca^{2+}), sodium (Na^+), and potassium (K^+) gradients across membranes, impairing excitation-contraction coupling.
- **Reduced blood flow and oxygen supply** further limit aerobic ATP production.

Symptoms:

- Weakness in contraction force.
- Slower responses.
- Burning sensation in the working muscle.
- Temporary inability to maintain activity.

Recovery:

Fatigue is usually reversible. Rest restores oxygen and ATP levels, while lactic acid is removed via the bloodstream and metabolized in the liver.

2. Muscle Cramps

Definition:

A muscle cramp is a **sudden, involuntary, and often painful contraction** of a muscle or muscle group, usually lasting from a few seconds to several minutes.

Causes:

- **Electrolyte imbalance:** Low levels of sodium, potassium, magnesium, or calcium interfere with normal neuromuscular transmission.
- **Overexertion or dehydration** during exercise or hot weather.



- Inadequate blood supply to working muscles.
- Nerve compression (e.g., in spinal disorders).

Symptoms:

- Sudden tightening and hardening of the muscle.
- Sharp pain and temporary immobility.

Relief and First Aid:

- Gentle stretching of the affected muscle.
- Massaging the area.
- Adequate hydration and electrolyte replacement.

3. Tetany

Definition:

Tetany is a condition characterized by **sustained, involuntary, and painful muscle contractions** due to increased excitability of nerves and muscles.

Causes and Mechanism:

- **Hypocalcemia (low blood calcium levels):** Calcium normally stabilizes the resting membrane potential of excitable cells. When calcium levels fall, sodium channels become overly active, causing uncontrolled nerve firing and continuous muscle contractions.
- **Parathyroid hormone deficiency** (hypoparathyroidism) can lead to tetany due to its role in regulating calcium balance.
- **Vitamin D deficiency** may also contribute.

Symptoms:

- Continuous, painful spasms of muscles.
- Tingling sensations in hands, feet, or face.
- Severe cases may cause laryngospasm (spasms of the laryngeal muscles), which can interfere with breathing and become life-threatening.

SCIENCE TITBITS

Unlike fatigue and cramps, tetany is not simply due to overwork of muscles—it is primarily linked to **calcium homeostasis**, demonstrating the vital role of minerals in neuromuscular function.

Treatment:

- Immediate calcium supplementation (oral or intravenous).
- Treatment of underlying cause (e.g., vitamin D deficiency, parathyroid disorder).

Summary Table

Condition	Definition	Major Cause(s)	Symptoms	Management
Fatigue	Decline in muscle force after prolonged activity	ATP depletion, lactic acid buildup, ion imbalance	Weakness, burning sensation	Rest, oxygen replenishment
Cramps	Sudden involuntary contraction	Electrolyte imbalance, dehydration, overuse	Sharp pain, temporary stiffness	Stretching, massage, hydration
Tetany	Sustained involuntary contraction	Low calcium (hypocalcemia), parathyroid disorders	Painful spasms, tingling, possible breathing difficulty	Calcium therapy, treating cause

SCIENCE, TECHNOLOGY AND SOCIETY CONNECTIONS

Justify why do the muscles pull but do not push.

Bones act as the levers, while joints perform as living fulcrums. Muscle, attached to bones by tendons and other connective tissue, exerts force by converting chemical energy into tension and contraction. When a muscle contracts, it shortens, in many cases pulling a bone like a lever across its hinge. Muscles move and by their motions we move. We are capable of performing a wide variety of actions, but despite this, muscle itself moves only by becoming shorter. They shorten and then they rest - in other words, a muscle can pull but it cannot push.



Reason out the rigor mortis.

When death occurs, ATP is no longer made. It is a short-lived chemical and so it runs out fairly quickly. This causes the muscles to lock into position as cross-bridges that formed between actin and myosin filaments before death can no longer be broken. The skeletal muscles undergo a partial contraction that causes the joints to become fixed. This condition, rigor mortis (rigidity of death), happens in all body muscles. It appears about four hours after death and lasts about 34 hours. After this time, muscles proteins are destroyed by enzymes within the cells and so rigor mortis disappears.

SLOs Based (MCQs)**MUSCLES**

1. What type of connective tissue joins bone to bone?
A. Tendon B. Ligament
C. Cartilage D. Fascia
2. Which term describes the fixed attachment point of a muscle?
A. Belly B. Tendon
C. Insertion D. Origin
3. The contractile part of a skeletal muscle is called:
A. Origin B. Insertion
C. Belly D. Joint
4. When a flexor muscle contracts, it:
A. Straightens the bone
B. Rotates the joint
C. Bends the bone at the joint
D. Fixes the joint in place
5. Which muscle group is primarily responsible for knee flexion?
A. Quadriceps femoris B. Gluteal muscles
C. Hamstring muscles D. Calf muscles
6. The quadriceps femoris muscle group is involved in:
A. Knee flexion B. Leg rotation
C. Hip flexion D. Knee extension
7. Which of the following is an example of antagonistic muscle action?
A. Biceps and triceps
B. Flexors and extensors
C. Hamstrings and quadriceps
D. All of the above
8. Which of these is a main flexor of the knee?
A. Biceps femoris B. Vastus lateralis
C. Rectus femoris D. Gastrocnemius
9. What is the role of semimembranosus muscle?
A. Knee extension B. Hip abduction
C. Knee flexion D. Ankle rotation
10. The quadriceps femoris muscle inserts at the:
A. Femur B. Tibia
C. Pelvis D. Patella
11. During muscle contraction, the myosin heads bind to:
A. Tropomyosin B. Calcium ions
C. G actin active sites D. Z line
12. Which structure gives thin filaments their twisted appearance?
A. Tropomyosin B. Myosin chain
C. F actin D. Sarcolemma
13. Which of the following statements is true about the sliding filament model?
A. Actin filaments shorten
B. Myosin filaments shorten
C. Filaments remain the same length but slide past one another
D. The sarcomere expands
14. Muscle contraction involves energy supplied by:
A. Glucose only B. Oxygen only
C. ATP D. Lactic acid
15. Which of the following best describes antagonistic muscles?
A. Muscles that contract together
B. Muscles with the same insertion
C. Muscles that perform opposite actions
D. Muscles controlled by different nerves
16. Which statement best describes the origin of biceps femoris?
A. It originates from tibia
B. It originates from fibula
C. It originates from femur and pelvic girdle
D. It originates from patella
17. The patella is enclosed in the tendon of which muscle group?
A. Hamstrings B. Gastrocnemius
C. Quadriceps femoris D. Adductors



18. What gives the actin filament its "pearl-like" appearance?
 A. Myosin chains B. G actin monomers
 C. Sarcoplasmic reticulum D. Troponin
- 21 19. Which muscle extends the leg at the knee joint?
 A. Semitendinosus B. Biceps femoris
20. In a relaxed muscle, which protein blocks the myosin-binding sites on actin?
 A. Troponin B. Tropomyosin
 C. Myoglobin D. Actomyosin

Answer Key

1. B	2. D	3. C	4. C	5. C	6. D	7. D	8. A	9. C	10. B
11. C	12. C	13. C	14. C	15. C	16. C	17. C	18. B	19. C	20. B

Short Questions

Q.1 Define the terms origin and insertion of a muscle.

Ans. The origin of a muscle refers to the fixed attachment point, usually located on a bone that remains stationary during muscle contraction. It serves as the anchor from which the muscle pulls. The insertion of a muscle is the attachment on the bone that moves when the muscle contracts. During contraction, the insertion is drawn towards the origin, producing motion at the joint. This arrangement provides mechanical advantage and enables coordinated movement. For example, in the biceps brachii, the origin is at the scapula while the insertion is on the radius. Understanding these points is essential in studying muscle actions and body mechanics.

MARKING SCHEME (3 Marks)

- Definition of origin – 1.5 marks; definition of insertion – 1.5 marks

Q.2 Explain the function of antagonistic muscles with an example.

Ans. Antagonistic muscles are muscle pairs that work in opposition to produce controlled and reversible movements at a joint. When one muscle contracts, the opposing muscle relaxes. For example, in bending the arm at the elbow, the biceps brachii (flexor) contracts while the triceps brachii (extensor) relaxes. Conversely, when straightening the arm, the triceps contracts and the biceps relaxes. This antagonistic relationship prevents uncontrolled motion, protects joints from injury, and ensures precision in body movements. Without such coordination, movement would be jerky and unstable. Thus, antagonistic muscles are essential for smooth locomotion, posture maintenance, and controlled motor activities in the body.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; function – 2 marks

Q.3 Describe the structure of actin filaments in skeletal muscles.

Ans. Actin filaments are a key component of the thin filaments in muscle fibers. They are composed of globular actin (G-actin) subunits, which polymerize into long chains called fibrous actin (F-actin). Two F-actin strands intertwine around each other, forming a helical, twisted double-stranded structure that looks like a string of pearls. Along these filaments, specific binding sites exist for myosin heads, which are exposed during contraction. Actin filaments also interact with regulatory proteins like tropomyosin and troponin, which control access to the myosin binding sites. This structural organization allows actin filaments to play a central role in the sliding filament model of muscle contraction.

MARKING SCHEME (3 Marks)

- G-actin – 1 mark; F-actin – 1 mark; twisted double strand – 1 mark

Q.4 Explain how the sliding filament model accounts for muscle contraction.

Ans. According to the sliding filament model, muscle contraction results from the interaction between actin (thin filaments) and myosin (thick filaments) within the sarcomere. During contraction, myosin heads attach to binding sites on actin, forming cross-bridges. Using energy from ATP hydrolysis, the myosin heads pivot and pull the actin filaments inward, causing them to slide past the myosin filaments. This action shortens the sarcomere, leading to overall muscle contraction. Importantly, the filaments themselves do not shorten; only their relative position changes. The coordinated sliding of millions of sarcomeres in muscle fibers produces the force and movement necessary for body activities.

MARKING SCHEME (3 Marks)

- Role of actin – 1.5 marks; role of myosin – 1.5 marks

Q.5 State the main muscles involved in knee joint flexion and extension.

Ans. The movement of the knee joint involves two major muscle groups. For flexion of the knee, the hamstring group—consisting of the biceps

MARKING SCHEME (3 Marks)

- Names of muscles – 1 mark; roles – 2 marks



femoris, semitendinosus, and semimembranosus—contracts to bend the leg backward. These muscles are located at the back of the thigh and provide strength during activities such as running or squatting. For **extension of the knee**, the **quadriceps femoris group**, located on the front of the thigh, contracts to straighten the leg. This group includes the rectus femoris, vastus lateralis, vastus medialis, and vastus intermedius. Together, these muscles ensure smooth, controlled flexion and extension necessary for walking, standing, and jumping.

Q.6 Describe the function of the quadriceps femoris muscle group.

Ans. The **quadriceps femoris** is a large muscle group located at the front of the thigh. It originates from the ilium and femur and inserts into the tibia through the quadriceps tendon, which encloses the patella. Its primary function is to **extend the leg at the knee joint**, making it essential for movements such as standing upright, walking, running, and jumping. It stabilizes the knee during weight-bearing activities and prevents the leg from collapsing when standing. Without its coordinated action, basic locomotor tasks like climbing stairs or rising from a sitting position would not be possible. The quadriceps is therefore one of the most powerful and functionally significant muscles of the lower limb.

MARKING SCHEME (3 Marks)

- Location – 1 mark; functions – 2 marks

Q.7 Explain the role of ATP in muscle contraction.

Ans. **ATP (adenosine triphosphate)** is the immediate source of energy for muscle contraction. During the sliding filament mechanism, ATP first binds to the myosin head, allowing it to detach from actin after a power stroke. Hydrolysis of ATP then re-cocks the myosin head into a high-energy state, ready for the next cycle. When the myosin head binds to actin again, the release of ADP and Pi generates the **power stroke**, pulling the actin filament inward. This continuous cycle, powered by ATP, shortens the sarcomere and produces contraction. Without ATP, muscles would remain stiff, as seen in rigor mortis. Thus, ATP is indispensable for contraction, relaxation, and prevention of fatigue.

MARKING SCHEME (3 Marks)

- Power stroke – 3 marks

Q.8 Explain why muscles work in pairs around a joint.

Ans. Muscles have the unique ability to **pull** when they contract, but they cannot **push** a bone back to its original position. Therefore, movements at a joint require **antagonistic muscle pairs**, where one muscle contracts while the opposing muscle relaxes. For example, at the elbow joint, the **biceps brachii** flexes the forearm while the **triceps brachii** extends it back. This arrangement ensures controlled, smooth, and reversible motion, preventing damage to joints. Antagonistic pairs also provide stability and maintain posture by balancing opposite forces. Without such coordination, movements would become uncontrolled and jerky. Hence, paired muscle action is crucial for precision and flexibility in human locomotion.

MARKING SCHEME (3 Marks)

- Definition of antagonistic movement – 2 marks; importance – 1 mark

Q.9 Describe the structure of myosin in thick filaments.

Ans. **Myosin** is the main protein of thick filaments in muscle fibers. Each myosin molecule has a long **tail** and two **globular heads**, which function as cross-bridges during contraction. The heads contain binding sites for actin and ATP, enabling them to attach, pivot, and release during the sliding filament cycle. About **300 myosin molecules** assemble together to form one thick filament. Their tails align in parallel to form the filament's shaft, while the heads project outward in all directions, ready to interact with surrounding actin filaments. This structural arrangement allows efficient force generation and coordinated shortening of the sarcomere, making myosin essential for muscle contraction.

MARKING SCHEME (3 Marks)

- Head, tail of myosin, binding site for actin, bundle of thick filaments – 3 marks

Q.10 Explain the significance of the knee joint's structure.

Ans. The knee joint, also known as the tibio-femoral joint, is a complex **hinge joint** formed between the femur, tibia, and patella. Structurally, it allows flexion and extension, but also limited rolling and gliding, giving it more versatility than a simple hinge. The joint is stabilized by strong **ligaments such as the anterior and posterior cruciate ligaments**, as well as the menisci, which cushion and distribute weight. Its structure enables both **stability for weight-bearing** and **flexibility for movement**, making it vital for activities such as walking, running, and squatting. Without its unique design, humans would not be able to balance efficiently or perform powerful lower limb movements.

MARKING SCHEME (3 Marks)

- Structure – 2 marks; relation of structure with function – 1 mark



I

Multiple Choice Questions (MCQs)

❖ Select the correct Answer.

1. Ribs originate from vertebral column at:
 - A. lumbar region
 - B. cervical region
 - C. thoracic region
 - D. pelvic region
2. Skeletal muscles contain dark band, which are anisotropic, are called
 - A. A band
 - B. I band
 - C. Z band
 - D. M line
3. The acetabulum provides the articular surface for the
 - A. humerus
 - B. femur
 - C. pelvis
 - D. fibula
4. Scapula is connected with sternum by
 - A. ribs
 - B. carpals
 - C. clavicle
 - D. atlas
5. Which statement correctly describes the smooth muscles?
 - A. Unstriated involuntary with spindle shape cells
 - B. Unstriated involuntary with multinucleate cells
 - C. Unstriated voluntary with uninucleate cells
 - D. Striated involuntary with spindle shape cell
6. Thin myofilaments consist of
 - A. actin, myosin, troponin
 - B. actin, tropomyosin, troponin
 - C. actin, tropomyosin, fibrin
 - D. actin, myoglobin, troponin
7. Which of the following changes occur when skeletal muscle contracts?
 - A. The A-bands shorten
 - B. The I-bands shorten
 - C. The Z-lines move further apart
 - D. The H-zone becomes more visible
8. A human internal organs are protected mainly by the
 - A. hydrostatic skeleton
 - B. axial skeleton
 - C. exoskeleton
 - D. appendicular skeleton
9. Arm and leg muscles are arranged in antagonistic pairs. How does this affect their functioning
 - A. it provides a backup if one of the muscles is injured
 - B. one muscle of the pair pushes while the other pulls
 - C. it allows the muscles to produce opposing movements
 - D. it doubles the strength of contraction
10. Which of the following bones in the human arm would correspond to the femur in the leg?
 - A. radius
 - B. ulna
 - C. tibia
 - D. humerus
11. The deep infolding of the muscle fibre membrane is called
 - A. sarcoplasmic reticula
 - B. Z lines
 - C. T-tubules
 - D. sarcomeres
12. Bone dissolving cells are called
 - A. chondrocytes
 - B. osteoblasts
 - C. osteoclasts
 - D. osteocytes
13. Which of the following cartilage is found at the end of long bones?
 - A. calcified
 - B. fibrous
 - C. elastic
 - D. hyaline
14. At times ligaments are overstretched or torn. It is called
 - A. sprain
 - B. dislocation
 - C. fracture
 - D. tension
15. Which ion is essential for muscle contraction?
 - A. Na
 - B. K
 - C. Ca
 - D. Cl
16. The tough layer of connective tissue surrounding each bone is called
 - A. tendon
 - B. ligament
 - C. periosteum
 - D. cartilage
17. The network of tubes that run through compact bone is called the
 - A. periosteum
 - B. joint
 - C. Haversian canals
 - D. marrow
18. Strips of tough connective tissue that hold bones together are known as
 - A. tendon
 - B. smooth muscles
 - C. Striated muscles
 - D. ligaments



19. Joints that allow circular movements are

- A. gliding joints
- B. ball-and-socket joints
- C. hinge joints
- D. pivot joints

20. Two proteins that are involved in the

- contraction of muscles are
- A. sarcomere and microfibril
- B. actin and myosin
- C. periosteum and cartilage
- D. ATP and acetylcholine

Answer Key

1. C	2. A	3. B	4. C	5. A	6. B	7. B	8. B	9. C	10. D
11. C	12. C	13. D	14. A	15. C	16. C	17. C	18. D	19. B	20. B

II

Short Answer Questions

Q.1 How do compact bone and spongy bone differ in structure?

Ans. Compact bone is the dense, solid form of bone that makes up the outer layer of most skeletal structures. It is composed of tightly packed **osteons (Haversian systems)**, each containing concentric lamellae arranged around a central canal with blood vessels and nerves. This organization provides rigidity, strength, and resistance to compression. In contrast, spongy bone, also called cancellous bone, has a lightweight, porous architecture made up of **trabeculae**. These trabeculae form a lattice that creates spaces filled with red bone marrow, supporting hematopoiesis. Spongy bone is found at the ends of long bones and in flat bones, where it reduces overall weight while still providing structural support. Thus, compact bone offers strength, while spongy bone provides flexibility and blood cell formation.

MARKING SCHEME (3 Marks)

- Compact bone – 1.5 marks; spongy bone – 1.5 marks

Q.2 Name three types of cells associated with bone and write their functions.

Ans. Bone tissue is maintained and remodeled by three main types of specialized cells. **Osteoblasts** are responsible for bone formation. They secrete osteoid, a protein-rich matrix, which later mineralizes with calcium salts to form hard bone tissue. **Osteoclasts** are large, multinucleated cells that break down bone by secreting acids and enzymes. This process, called bone resorption, releases calcium and phosphate into the bloodstream and is vital for bone remodeling. **Osteocytes** are mature bone cells that originate from osteoblasts once they become embedded in the matrix. They reside in lacunae and maintain bone health by regulating mineral balance and communicating with other cells via canaliculi. Together, these cells ensure growth, repair, and continuous renewal of bone tissue.

MARKING SCHEME (3 Marks)

- 1 mark each for osteoblasts, osteoclasts, and osteocytes

Q.3 Compare the structure of bone with that of cartilage.

Ans. Bone and cartilage are both connective tissues but differ greatly in structure and function. **Bone** is rigid, mineralized, and vascular. Its hardness comes from calcium salts such as calcium phosphate and calcium carbonate, embedded in a collagen fiber framework. Bones contain osteocytes in lacunae, organized into osteons, and have a rich blood supply, which allows rapid repair and remodeling. In contrast, **cartilage** is flexible, non-mineralized, and avascular. It has a gel-like extracellular matrix made of collagen and elastic fibers, giving it resilience and flexibility. The cells, called chondrocytes, are located in lacunae and receive nutrients by diffusion. Thus, while bone provides structural strength and protection, cartilage cushions joints, supports structures, and allows flexibility.

MARKING SCHEME (3 Marks)

- Bone structure – 1.5 marks; cartilage structure – 1.5 marks

Q.4 Name the bones of the axial and appendicular skeleton.

Ans. The human skeleton is divided into two major parts: **axial skeleton** and **appendicular skeleton**. The axial skeleton forms the central framework of the body. It includes the **skull (cranium and facial bones)**, the **vertebral column (33 vertebrae)**, and the **rib cage (sternum and ribs)**. This division provides support, posture, and protection for vital organs such as the brain, spinal cord, heart, and lungs. The appendicular skeleton consists of the bones of the limbs and girdles. It includes the **pectoral**

MARKING SCHEME (3 Marks)

- Axial skeleton – 1.5 marks; appendicular skeleton – 1.5 marks

girdle (clavicle and scapula), **upper limbs** (humerus, radius, ulna, carpals, metacarpals, phalanges), **pelvic girdle** (ilium, ischium, pubis), and **lower limbs** (femur, tibia, fibula, tarsals, metatarsals, phalanges). Together, these bones allow locomotion and manipulation of the environment.

Q.5 Name the bones of the cranium.

Ans. The cranium forms the upper part of the skull and encloses the brain, providing protection and shape to the head. It is composed of **eight bones**, which are classified into paired and unpaired groups. The paired bones include the **parietal bones (2)** and the **temporal bones (2)**. These form the sides and roof of the cranial cavity. The unpaired bones include the **frontal bone**, which forms the forehead, the **occipital bone**, which forms the back and base of the skull, the **sphenoid bone**, which lies at the base of the cranium and connects with all other cranial bones, and the **ethmoid bone**, located between the eyes and contributing to the nasal cavity. Collectively, these bones protect the brain and support sensory structures.

MARKING SCHEME (3 Marks)

- Paired bones – 2 marks; unpaired bones – 1 mark

Q.6 Describe the five groups of vertebrae.

Ans. The vertebral column is the central supporting structure of the skeleton and is divided into **five groups** of vertebrae, totaling **33 in number**. The **cervical vertebrae (7)** form the neck region, supporting the skull and allowing head movement. The **thoracic vertebrae (12)** are attached to the ribs and make up the mid-back, contributing to the rib cage. The **lumbar vertebrae (5)** are the largest and strongest, supporting most of the body's weight in the lower back. The **sacral vertebrae (5)** are fused together to form the sacrum, connecting the spine to the pelvis. Finally, the **coccygeal vertebrae (4)** fuse to form the coccyx or tailbone, which provides attachment to ligaments and small muscles.

MARKING SCHEME (3 Marks)

- Cervical, thoracic, lumbar – 1.5 marks; sacral, coccygeal – 1.5 marks

Q.7 What is the structure of the human rib cage?

Ans. The rib cage is a bony framework that protects vital organs such as the heart and lungs while assisting in breathing. It is made up of **12 pairs of ribs**, the **sternum**, and the **thoracic vertebrae**. The ribs are divided into three categories. The **true ribs (first 7 pairs)** are directly attached to the sternum through costal cartilages, providing firm protection. The **false ribs (next 3 pairs, 8th–10th)** are attached indirectly to the sternum via cartilage of the 7th rib, offering partial flexibility. The **floating ribs (last 2 pairs, 11th and 12th)** are not attached to the sternum at all, ending freely in the abdominal wall. Together, the rib cage provides both rigidity and elasticity, ensuring effective respiration.

MARKING SCHEME (3 Marks)

- Number of ribs – 1 mark; types of ribs – 2 marks

Q.8 Name the bones that form (a) the pectoral girdle and (b) the pelvic girdle.

Ans. The human skeleton contains two major girdles that connect the limbs to the axial skeleton. **(a) The pectoral girdle** consists of two bones: the **clavicle (collarbone)** and the **scapula (shoulder blade)**. These bones support the upper limbs and allow wide ranges of motion at the shoulder joint. **(b) The pelvic girdle** is formed by the fusion of three bones on each side: the **ilium, ischium, and pubis**. Together, they form the hip bone, which unites with its counterpart at the pubic symphysis. The pelvic girdle provides strong support for the lower limbs, bears the weight of the body, and protects pelvic organs such as the bladder and reproductive structures.

MARKING SCHEME (3 Marks)

- Pectoral girdle – 1.5 marks; pelvic girdle – 1.5 marks

Q.9 Name the bones of the upper limb and lower limb.

Ans. The limbs of the human skeleton are specialized for mobility and support. The **upper limb** consists of the **humerus (upper arm)**, **radius and ulna (forearm)**, **carpals (8 wrist bones)**, **metacarpals (5 palm bones)**, and **phalanges (14 finger bones)**. These bones provide flexibility for tasks such as grasping, lifting, and writing. The **lower limb** includes the **femur (thigh bone, the longest bone in the body)**, **tibia and fibula (leg bones)**, **patella (kneecap)**, **tarsals (7 ankle bones)**, **metatarsals (5 foot bones)**, and **phalanges (14 toe bones)**. These bones are adapted for weight-bearing, balance, and locomotion, making them stronger and larger compared to the bones of the upper limb.

MARKING SCHEME (3 Marks)

- Upper limb bones – 1.5 marks; lower limb bones – 1.5 marks

Q.10 What are the main types of joints found in bones (human skeleton)?

Ans. Joints are structural connections between bones that allow varying degrees of movement. They are classified into three major types.



Fibrous joints are immovable, where bones are held together by dense connective tissue, such as the sutures of the skull. **Cartilaginous joints** are slightly movable, where bones are connected by cartilage, such as intervertebral discs and the pubic symphysis. **Synovial joints** are freely movable and the most common type in the body. They contain a synovial cavity filled with lubricating synovial fluid, and articular surfaces are covered by cartilage. Synovial joints are further divided into subtypes, such as **hinge joints (knee, elbow)**, **ball-and-socket joints (shoulder, hip)**, **pivot joints (atlas-axis)**, **gliding joints (wrist)**, and **saddle joints (thumb)**.

MARKING SCHEME (3 Marks)

- Fibrous – 1 mark; cartilaginous – 1 mark; synovial – 1 mark

Q.11 What is fibrous joint?

Ans. A fibrous joint is an **immovable type of joint** where bones are directly connected by **dense connective tissue rich in collagen fibers**, without a joint cavity. This structure ensures stability but does not allow movement. Fibrous joints are important in protecting delicate organs, such as the brain, by keeping bones firmly in place. The best example is the **sutures of the skull**, where cranial bones are tightly interlocked and fused, ensuring the skull remains strong and rigid. Another type of fibrous joint is the **syndesmosis**, where bones are joined by a ligament, such as between the tibia and fibula. Overall, fibrous joints emphasize protection and strength over mobility.

MARKING SCHEME (3 Marks)

- Definition – 1.5 marks; example – 1.5 marks

Q.12 What are the four steps required for bone fracture repair?

Ans. The healing of a bone fracture is a **complex, stepwise process** that restores both structure and function.

1. **Hematoma formation:** Immediately after fracture, blood vessels rupture, forming a clot that stabilizes the area and initiates inflammation.
2. **Fibrocartilaginous callus formation:** Fibroblasts and chondroblasts invade, producing collagen and cartilage that temporarily bridge the broken ends.
3. **Bony callus formation:** Osteoblasts gradually replace the soft callus with a hard bony callus made of spongy bone.
4. **Bone remodeling:** Over weeks to months, osteoclasts and osteoblasts reshape the bone, replacing spongy bone with compact bone, restoring the bone to near-original shape and strength.

MARKING SCHEME (3 Marks)

- All four steps correctly explained – 3

Q.13 What skeletal structures are affected from the osteoarthritis?

Ans. Osteoarthritis is a **degenerative joint disorder** that primarily affects **synovial joints**. The disease begins with the gradual breakdown of **articular cartilage**, which normally cushions and allows smooth movement of bones. As cartilage erodes, the joint space narrows, and bones may rub against each other, causing pain and stiffness. In response, bones may **develop osteophytes (bone spurs)** at joint margins. Additionally, changes occur in the underlying **subchondral bone**, which becomes denser and less elastic. The surrounding ligaments and joint capsule may also thicken, further restricting mobility. Commonly affected joints include the **knees, hips, hands, and spine**. Thus, osteoarthritis disrupts multiple skeletal structures, ultimately impairing normal joint function.

MARKING SCHEME (3 Marks)

- Cartilage affected – 1 mark; other structures and symptoms – 2 marks

Q.14 List the major parts of skeletal muscle fiber and write the function of each part.

Ans. A skeletal muscle fiber is a **specialized, multinucleated cell** designed for contraction. Its main parts include:

- **Sarcolemma:** The cell membrane that conducts electrical impulses (action potentials) to trigger contraction.
- **Sarcoplasm:** The cytoplasm containing glycogen, mitochondria, and myoglobin to provide energy.
- **Myofibrils:** Long contractile elements composed of actin and myosin filaments, responsible for generating force.
- **Sarcoplasmic reticulum (SR):** A network that stores calcium ions and releases them when contraction is initiated.
- **T-tubules:** Invaginations of the sarcolemma that transmit action potentials deep into the fiber, ensuring coordinated contraction.
- Together, these structures allow skeletal muscles to contract efficiently, respond rapidly to stimuli, and sustain body movements.

MARKING SCHEME (3 Marks)

- Naming parts – 1.5 marks; functions – 1.5 marks



Q.15 What is Z line and M line and what are their functions?

Ans. The sarcomere is the functional unit of a muscle fiber, and two important structural features within it are the Z line and M line. The Z line marks the boundary between adjacent sarcomeres and serves as the anchoring site for thin filaments (actin). It maintains the structural integrity of the sarcomere during contraction and relaxation. The M line, located in the center of the sarcomere within the H zone, anchors the thick filaments (myosin) and keeps them aligned during contraction. Together, the Z line and M line ensure proper filament organization, efficient sliding of actin and myosin, and overall stability of the contractile process.

MARKING SCHEME (3 Mark)

- Z line function – 1.5 marks; M line function – 1.5 marks

Q.16 How the arrangement of actin filaments and myosin filaments produce I band, A band, and H zone?

Ans. The microscopic arrangement of actin (thin) and myosin (thick) filaments within a sarcomere creates distinct regions visible under a microscope. The I band appears as a light region because it contains only thin filaments (actin), anchored at the Z line. The A band is darker and corresponds to the entire length of the thick filaments (myosin). Within the A band, there is overlap where actin and myosin interact during contraction. At the center of the A band lies the H zone, which contains only myosin filaments without overlap of actin. This structural arrangement is fundamental to the sliding filament mechanism of muscle contraction, where bands shorten or disappear during contraction, but filaments themselves remain the same length.

MARKING SCHEME (3 Marks)

- I band – 1 mark; A band – 1 mark; H zone – 1 mark

Q.17 Describe the antagonistic arrangement of skeletal muscles.

Ans. Skeletal muscles are arranged in antagonistic pairs because a single muscle can only pull when contracting, not push. In this arrangement, one muscle contracts to produce movement in one direction, while the opposing muscle relaxes. When movement needs to be reversed, the roles switch. For example, at the knee joint, the quadriceps femoris contracts to extend the leg, while the hamstring group relaxes. Conversely, when the hamstrings contract to flex the knee, the quadriceps relax. This coordination ensures smooth, controlled, and reversible movements. Antagonistic action also prevents damage to joints by balancing forces, allowing stability. Thus, antagonistic muscle pairs are essential for efficient locomotion and postural control in the human body.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; example and explanation – 2 marks

Q.18 Why are ligaments elastic and why does the tendon need to be inelastic.

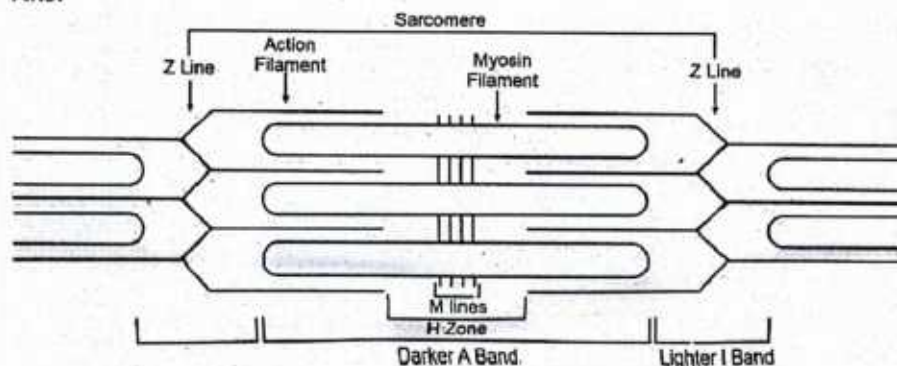
Ans. Ligaments and tendons are both connective tissues, but they differ in function and structure. Ligaments connect bone to bone at joints. They are elastic because limited stretching allows flexibility and controlled movement, while still preventing excessive displacement of bones that could damage the joint. Their elasticity ensures stability combined with mobility. In contrast, tendons connect muscle to bone. They must be inelastic to transmit the full force of muscle contraction directly to bones without energy loss due to stretching. If tendons were elastic, movements would become weak, imprecise, and delayed. Therefore, ligaments provide flexibility and joint stability, while tendons provide strength and efficiency in force transfer, ensuring precise and powerful skeletal movements.

MARKING SCHEME (3 Marks)

- Ligaments – 1.5 marks; tendons – 1.5 marks

Q.19 Draw a diagram of sarcomere and label its parts.

Ans.



MARKING SCHEME (3 Marks)

- Correct outline of sarcomere with Z lines and filaments – 1 mark
- Accurate labeling of major parts (Z line, M line, I band, A band, H zone) – 1 mark
- Neatness and correct positioning of actin & myosin filaments – 1 mark

Q.20 Write the difference between:

(a) Epiphysis and diaphysis for a long bone

Ans. The epiphysis refers to the rounded ends of a long bone, found at both the proximal and distal ends. It is made up mainly of **spongy bone** that contains red bone marrow, which is important for **hematopoiesis** (blood cell production). Epiphyses are covered by articular cartilage and participate in **joint formation**, helping bones move smoothly against each other. In contrast, the diaphysis is the long, cylindrical shaft of the bone located between the two epiphyses. It consists mostly of **compact bone** surrounding a hollow medullary cavity, which usually contains yellow marrow for fat storage. Its primary role is to provide **strength, leverage, and weight-bearing support** for the skeleton.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; composition – 1 mark; function/location – 1 mark

(b) Periosteum and Endosteum

Ans. The periosteum is a tough, dense, fibrous membrane that covers the outer surface of bones except at articular cartilage. It consists of two layers: an **outer fibrous layer** that protects the bone and allows attachment of ligaments and tendons, and an **inner osteogenic layer** containing osteoblasts, which aid in bone growth and repair. On the other hand, the endosteum is a thin, delicate, vascular membrane lining the inner surface of the bone's **medullary cavity**. It contains **osteoblasts and osteoclasts**, which actively participate in **bone growth, remodeling, and repair**. While the periosteum is external and provides mechanical protection, the endosteum is internal and directly regulates bone turnover from within.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; composition – 1 mark; function/location – 1 mark

(c) Compact bone and Spongy bone

Ans. **Compact bone** is the dense, tightly packed type of bone tissue forming the **outer layer** of all bones. Its structure is based on repeating cylindrical units called **osteons or Haversian systems**, which contain blood vessels and nerves, making it **highly vascular**. Its function is to provide **strength, rigidity, and resistance to stress**, ensuring structural support for the body. In contrast, **spongy bone** (cancellous bone) has a porous, lattice-like network of **trabeculae**. It is found mainly at the **ends of long bones** and inside flat bones such as the sternum. Spongy bone reduces the overall weight of bones while also housing **red marrow**, which is essential for blood cell production.

MARKING SCHEME (3 Marks)

- Structure – 1 mark; location – 1 mark; function/vascularity – 1 mark

(d) Axial skeleton and Appendicular skeleton

Ans. The **axial skeleton** consists of the bones along the **central axis** of the body, including the skull, vertebral column, sternum, and ribs. It is made up of **80 bones** and its primary function is to provide **support and protection** for vital organs such as the brain, heart, and lungs. The **appendicular skeleton**, on the other hand, consists of **126 bones**, including those of the **limbs** as well as the pectoral (shoulder) and pelvic (hip) girdles. Its main function is to facilitate **movement, locomotion, and manipulation of the environment**. Thus, while the axial skeleton maintains body posture and protection, the appendicular skeleton provides mobility and interaction with surroundings.

MARKING SCHEME (3 Marks)

- Components – 1 mark; function – 1 mark; number/location – 1 mark

(e) True ribs and False ribs

Ans. **True ribs** are the first seven pairs (1–7) of ribs, also known as **vertebrosternal ribs**. They are directly attached to the sternum through their own costal cartilages. Their function is to provide **firm protection** to the heart, lungs, and major blood vessels by forming a strong thoracic cage. In contrast, **false ribs** are the last five pairs (8–12). The 8th to 10th pairs, called **vertebrochondral ribs**, attach indirectly to the sternum via the cartilage of the 7th rib, while the 11th and 12th pairs, called **floating ribs**, have no anterior attachment. False ribs provide **flexibility** to the lower thoracic cage and allow for expansion during breathing movements.

MARKING SCHEME (3 Marks)

- Attachment – 1 mark; pairs – 1 mark; function/alternate name – 1 mark

(f) False ribs and Floating ribs

Ans. **False ribs** are the 8th, 9th, and 10th pairs of ribs, also called **vertebrochondral ribs**. They do not attach directly to the sternum but instead connect through the cartilage of the 7th rib. Their function is to contribute to the **flexibility and expansion of the rib cage** during

MARKING SCHEME (3 Marks)

- Attachment – 1 mark; pairs – 1 mark; function/type – 1 mark



breathing. In contrast, **floating ribs** are the 11th and 12th pairs, also known as **vertebral ribs**. These ribs are not attached to the sternum at all; they are only connected posteriorly to the thoracic vertebrae. Floating ribs mainly help protect the **kidneys** and allow for greater flexibility and torsional movement of the torso. Both play important roles in thoracic function but differ in structure and attachment.

(g) Atlas (C1) and Axis (C2) vertebrae

Ans. The **atlas (C1)** is the first cervical vertebra and has a **ring-like structure** without a body. It articulates superiorly with the **occipital condyles** of the skull, supporting the head, and inferiorly with the axis. Its primary function is to allow the **nodding motion** of the head (as in "yes"). The **axis (C2)**, on the other hand, is the second cervical vertebra. It is characterized by a unique upward projection called the **dens or odontoid process**, which acts as a pivot for rotation of the atlas and skull. This enables the **side-to-side movement** of the head (as in "no"). Together, atlas and axis form a specialized joint essential for head mobility.

MARKING SCHEME (3 Marks)

- Structure – 1 mark; function – 1 mark; articulation/location – 1 mark

(h) Rheumatoid arthritis and Gouty arthritis

Ans. **Rheumatoid arthritis (RA)** is an autoimmune disorder in which the body's immune system attacks the **synovial membrane**, causing chronic inflammation. Symptoms include **joint stiffness, swelling, deformity, and fatigue**, often affecting multiple small joints of the hands and feet. It is **systemic** and may involve other organs. In contrast, **gouty arthritis** is caused by **accumulation of uric acid crystals** in joints. It is characterized by **sudden, severe pain, redness, swelling, and warmth**, commonly in the big toe, knees, or ankles. Unlike RA, it is usually **localized** and not systemic. While RA is long-term and progressive, gout often occurs in acute attacks triggered by diet, metabolism, or kidney inefficiency.

MARKING SCHEME (3 Marks)

- Cause – 1 mark; symptoms – 1 mark; common sites/nature – 1 mark

(i) Simple fracture and Compound fracture

Ans. A **simple fracture** refers to a bone break in which the bone does not pierce the skin. It remains an **internal injury**, with minimal risk of infection, and is usually treated by immobilization with a cast or splint to allow natural healing. In contrast, a **compound fracture** involves a bone break where the broken end protrudes through the skin. This creates an **open wound**, visible externally, and carries a **high risk of infection** due to exposure to bacteria. Treatment is more complicated, often requiring **surgical intervention, alignment, and sterilization** to prevent complications. While both are fractures, compound fractures are more dangerous due to their severity and potential complications.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; risk/visibility – 1 mark; treatment – 1 mark

(j) Tropomyosin and Troponin

Ans. **Tropomyosin** is a rod-shaped protein that winds around the actin filament, covering the **myosin-binding sites** in a resting muscle. Its primary role is to **block contraction** by preventing myosin from attaching to actin. **Troponin**, on the other hand, is a **globular protein complex** attached to tropomyosin at intervals along the actin filament. It has **calcium-binding sites** and acts as a regulatory switch. When calcium ions bind to troponin, it undergoes a conformational change that shifts tropomyosin, thereby **exposing the binding sites on actin**. This enables myosin heads to attach and initiate contraction. Thus, tropomyosin maintains inhibition at rest, while troponin regulates contraction through calcium sensitivity.

MARKING SCHEME (3 Marks)

- Structure – 1 mark; function – 1 mark; role in contraction/calcium binding – 1 mark

(k) Tendons and Ligaments

Ans. **Tendons** are strong bands of connective tissue that attach **muscle to bone**. They are made of **dense regular connective tissue** arranged in parallel bundles, which makes them tough and resistant to stretching. Because tendons are **inelastic**, they can transmit the full pulling force generated by muscles directly to bones, enabling precise and powerful movement. **Ligaments**, however, connect **bone to bone** at joints. They also consist of dense regular connective tissue, but with more **elastic fibers**, which provide limited flexibility. Their main function is to **stabilize joints**, prevent dislocation, and restrict excessive or abnormal movement. Thus, tendons facilitate **movement transmission**, while ligaments provide **joint stability** and controlled flexibility.

MARKING SCHEME (3 Marks)

- Definition – 1 mark; elasticity – 1 mark; function/composition – 1 mark

(l) Callus and Bony callus in Bone Repair

Ans. During bone healing, a **callus** is the initial soft bridge that forms between the broken bone ends. It consists mainly of **fibrocartilaginous tissue** and appears during the **intermediate stage** of healing. Its role is to **stabilize the fracture site** and provide a temporary scaffold for new bone formation. Later, the callus is gradually replaced by a **bony callus**, which is made of **spongy bone tissue**. The bony callus is stronger and appears in the **later stage** of healing, helping to restore the continuity of the bone. Eventually, this spongy bone is remodeled into compact bone, completing the repair. Thus, the callus begins stabilization, while the bony callus restores structural strength.

MARKING SCHEME (3 Marks)
• Definition – 1 mark; composition /stage – 1 mark; function – 1 mark

(m) Tetany and Tetanus

Ans. Tetany is a medical condition caused by **low blood calcium levels (hypocalcemia)**. This imbalance increases **neuromuscular excitability**, leading to involuntary **sustained muscle spasms, cramps, and twitching**. Tetany is not infectious and is usually treated by correcting calcium levels through **supplements or intravenous calcium**. In contrast, **tetanus** is an infectious disease caused by the bacterium **Clostridium tetani**, which produces a powerful toxin. The toxin blocks **inhibitory neurotransmitters** in the nervous system, causing severe **muscle stiffness and spasms**, often starting in the jaw ("lockjaw") and spreading to the body. Treatment involves administration of **antitoxin, antibiotics, and muscle relaxants**. Hence, tetany arises from electrolyte imbalance, whereas tetanus results from bacterial infection.

MARKING SCHEME (3 Marks)
• Cause – 1 mark; symptoms / mechanism – 1 mark; treatment – 1 mark



Long Answer Questions

Q.1 Explain the structure of bone with diagram.

Ans. Please See SLO [B-12-R-61]

Q.2 Explain the structure of cartilage with diagram.

Ans. Please See SLO [B-12-R-61]

Q.3 Give a detail account of bone repair.

Ans. Please See SLO [B-12-R-66]

Q.4 Describe three types of muscle tissues in man.

Ans. Please See SLO [B-12-R-68]

Q.5 Explain the ultra structure of skeletal muscle.

Ans. Please See SLO [B-12-R-69]

Q.6 Explain the sliding filament model of muscle contraction.

Ans. Please See SLO [B-12-R-70]

Q.7 Explain the action of antagonistic muscles in the movement of knee joint in man.

Ans. Please See SLO [B-12-R-71]

Q.8 Describe the following muscle disorders:

a. Muscle fatigue b. Cramp c. Tetany

Ans. Please See SLO [B-12-R-72]

